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Landmark-Focused vs Turn-by-Turn Navigation in Minecraft Virtual Environments

An Exploratory Investigation of Interface Design, Task Complexity, and Spatial Knowledge Acquisition

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Abstract

This study explores the effect of two different navigation interfaces, a minimalist *landmark-focused (LF)* and a more aided *turn-by-turn (TBT)*, on spatial knowledge acquisition and navigation performance in a custom-designed Minecraft virtual environment. The turn-by-turn interface is designed to represent a familiar navigation system, similar to commonly used smartphone and in-car applications, showing directional cues (e.g., "go straight, then turn left") and a path line to follow. The landmark-focused interface aims to encourage a different approach than passive following, stimulating participants to use landmarks to orient and encode their surroundings. Four forth-and-back navigation tasks with different levels of challenge were designed. The effects of interface and task difficulty were measured using navigation performance (travel time) and configurational knowledge (landmark placement error, pointing error and time) variables and via linear mixed-model analyses. Eye-tracking data was collected along with self-reported measures of sense of direction, spatial orientation, and video games experience questionnaires. We found that interface type has an impact on travel time, but no significant effect on survey knowledge measures. Our results show that task difficulty is a stronger predictor of performance across variables, and suggest that individual differences play an important role as well.

Table of Contents

1. Introduction

Navigation is a fundamental human ability that supports our everyday interactions with the environment. We constantly interact with our surroundings, commuting to work, driving to the gym, or walking to the closest shop. Although navigating often appears effortless, moving through an environment is a critical skill that relies on a set of cognitive processes: visualising one's location and orientation in space, understanding where the destination is located, selecting a sequence of actions to reach it, and, finally, executing the planned route (Ishikawa et al., 2008).

Cognitive research commonly distinguishes between three forms of spatial knowledge. *Landmark knowledge* encodes relevant objects or landscape features that serve as anchor reference points. *Route knowledge* stores sequences of segments connecting those landmarks, as a succession of decision points to travel through. Finally, *survey knowledge* is a more configurational, map-like representation that captures metric relationships between places, providing leverage to plan novel routes and shortcuts (Richardson et al., 1999; Siegel & White, 1975).

When navigating a new environment, we can use two strategies relying on different brain systems depending on the situation. One strategy is to construct and use a *cognitive map* of the environment (Tolman, 1948), encoding landmark locations and the routes interconnecting them; this proves especially useful when navigating new environments or in contexts where adaptability is needed. The alternative is a *stimulus-response* strategy, which works by associating motor responses (e.g., turning left) with specific stimuli or locations; it relies on a brain region associated with habit learning and enables us to store habitual routes we navigate in autopilot (Dahmani & Bohbot, 2020).

1.1. Navigation aids

A *navigation aid* is any medium or interface that assists wayfinding and spatial orientation by providing information. Such aids range from paper maps and verbal directions to software interfaces; they vary in modality (visual, auditory, multimodal) and in how they help navigation.

Historically, paper maps have served as an external representation of spatial knowledge. Using a map facilitates learning of global survey knowledge, although it tends to be closely tied to the map orientation (Richardson et al., 1999).

The introduction of consumer digital navigation systems, first as standalone GPS units and later as smartphone apps and in-car interfaces, has profoundly changed how people navigate. These devices, initially only working offline with maps downloaded separately, improved over the years. Thanks to advances in technology, they are now powerful and accurate, enabling every smartphone to access high-resolution satellite images, street-level views, and real-time traffic data. Such interfaces improve immediate efficiency, helping users reach their destinations without worrying about route planning. However, they also raise the question about what users learn (or fail to learn) while being guided.

Although the effect size depends on interface features, task demands, and user characteristics, growing literature suggests that use of turn-by-turn assisted navigation, especially when habitual and prolonged, has a detrimental effect on spatial knowledge acquisition and navigation performance (Ben-Elia, 2021; Dahmani & Bohbot, 2020; Gardony et al., 2015). These interfaces are not designed to provide a traditional cognitive experience during navigation; instead, they are precisely studied to enhance navigation efficiency (Dey et al., 2018). Following prompts encourages a passive approach to navigation, as users no longer need to put active effort into encoding their surroundings and planning routes (Huston & Hamburger, 2023). Various explanations were proposed, among them attention division between the device and the environment (Gardony et al., 2015) and the promotion of stimulus-response navigation strategies (Dahmani & Bohbot, 2020).

Recent research advocates for interface designs that balance navigational efficiency with incidental spatial learning, for example, emphasising the role of landmarks as attention-guiding cues (Ben-Elia, 2021; Kapaj et al., 2024; Liu et al., 2022); this supports the idea that navigation aids do not simply help or harm, but can instead be thoughtfully designed to promote learning (Dey et al., 2018).

1.2. Virtual Environments

Video games can serve as a controlled and rich testbed for studying cognitive processes and behaviours. An experimental setup that requires complex laboratory organisation can be emulated virtually, reducing setup time and allowing for quicker prototyping. In these simulated environments, users can perform virtual tasks that closely approximate real-world ones, enabling the collection of large amounts of data at a faster pace. Desktop eye-trackers that capture at high frequency are ideally suited for this kind of experiment, although new wearable devices are now being commercialised. Furthermore, experiments can be replicated more easily. Many other games have been studied for other reasons, such as Tetris (Lau-Zhu et al., 2017) or Starcraft II (Nieciecka et al., 2025). In the field of spatial navigation, popular game engines such as Unity3D have been used for building simulations (Carbonell-Carrera et al., 2020; Hejtmánek et al., 2018).

Minecraft is an open-ended, 3D cube-based game (i.e., where the entire world consists of cubes) that gained popularity over the past decade. Simulated physics and simple movement controls make it intuitive for new players to pick up. Finally, it not only offers built-in in-game commands to construct complex tasks, but also provides an extensive set of APIs for modding. A broad range of tasks can be pursued and studied, spanning from navigation (Simon et al., 2022) to spatial abilities assessment (Peters et al., 2021). These are a few reasons why it is an interesting and versatile experimental platform in the context of virtual environments.

1.3. Study aims and hypotheses

This thesis leverages a custom Minecraft environment to investigate whether a *landmark-focused (LF)* navigation interface facilitates spatial knowledge acquisition and navigation performance compared to a conventional *turn-by-turn (TBT)* interface. The TBT interface aims to represent a familiar, dominant commercial model, showing directional cues (e.g., "go straight, then turn left") and a guiding line on the map. The LF interface proposes a different approach to wayfinding, encouraging participants to make use of landmarks to orient themselves and encode their surroundings.

Following a within-subject counterbalanced design, participants will be tested across four tasks spanning different difficulty levels that require forth-and-back navigation between two points. Tasks consist of two consecutive phases, similar to other studies (Hejtmánek et al., 2018): a *learning phase*, where players are provided with a navigation interface and must reach the end goal, and a *recall phase*, where their navigation interface is hidden, and they are asked to return to their starting point. After each phase, subjects are asked to point towards their point of origin. Recorded measures include completion time, pointing angle error, and decision time, continuous landmark-placement error, gaze fixation metrics, and self-reported spatial skill measures such as sense of direction and mental rotation abilities.

Research Question. *How do landmark-focused (LF) and turn-by-turn (TBT) navigation interfaces differ in their effects on spatial knowledge acquisition and navigation performance within Minecraft virtual environments?*

Two main hypotheses follow:

Hypothesis 1 (spatial knowledge and navigation). The LF interface will yield more accurate spatial survey knowledge (lower angular error and decision time in pointing, higher landmark-placement accuracy) and a shorter travel time in unaided return navigation compared to TBT.

We expect landmark-focused guidance to encourage a more active encoding of the environment compared to turn-by-turn, leveraging landmarks during spatial knowledge acquisition; research shows they serve as cognitive anchors to determine the orientation of cognitive maps and encode metric distances between locations (Epstein et al., 2017), and help in building spatial representations of the environment (Kapaj et al., 2024; Liu et al., 2022).
On the other hand, turn-by-turn GPS systems are a great aid for navigation efficiency, which we expect to see in the learning phase travel time; nonetheless, they have been linked with cognitive offloading and active spatial processing impairment (Dey et al., 2018; Gardony et al., 2015). Research with a similar experimental design observed that TBT players showed poorer spatial memory upon aid removal in the recall phase (Hejtmánek et al., 2018).

Hypothesis 2 (eye-tracking, exploratory). The TBT and LF interfaces will produce different gaze patterns. We expect TBT users to show more frequent fixations on the navigation interface and fewer on the environment, while LF users show higher fixation frequency on the environment.

Prior eye-tracking work supports more frequent glances at the navigation device but shorter environmental fixations in GPS users (Hejtmánek et al., 2018), along with TBT causing distraction from the environment (Gardony et al., 2015). This hypothesis remained exploratory due to limited time, and its results are discussed accordingly.

2. Background

In this section, we explore the existing literature in the field of spatial knowledge, spatial navigation, and virtual environments, providing a foundation for our research.

2.1. Spatial Knowledge

The collection of information about space, its locations, and the relations between them is what we call *spatial knowledge*. It is fundamental for individuals to accomplish spatial tasks, such as locating themselves, planning routes, and moving in an environment (Dey et al., 2018). This internal representation is stored in the brain, in the *spatial memory*, and is constantly updated as we navigate, providing a more accurate representation of our environment; it is a compact collection of knowledge, containing the spatial organisation of places, their relative distances, and hierarchical arrangement (Ahmadpoor & Shahab, 2019). The process by which we collect this set of information is called *spatial knowledge acquisition*.

2.1.1. Classical literature: cognitive maps and place cells

Classical literature introduces the concept of *cognitive maps*, defined as a mental representation of the layout of one's environment (Tolman, 1948). Tolman's research sought to explain rats' behaviour in a navigational task, observing that when the known route was blocked, the rodent would quickly switch to a more direct path to reach its goal. Years later, researchers identified part of the neural structure responsible for such behaviour, which became known as *place cells* (J & J, 1971). These cells, located in the hippocampus, an area of the brain responsible for memory, navigation, and cognition (Lisman et al., 2017), play a crucial role in the construction of cognitive maps. They enable the brain to build a positional coordinate system, as cells fire more vigorously when the user is in a specific location, allowing the encoding of landmarks. Vast advancements in research have been made in the following years, bringing to the discovery of more position-related cells in the brain, such as *grid cells* (Hafting et al., 2005), providing encoding of distances, and *head navigation cells* (Taube et al., 1990), used to encode orientation; together, they give us a better sense of how the brain understands its location and the environment it is in. Further studies, using functional imaging, found that navigation tasks in *virtual environments (VE)* show neuronal activity patterns similar to those in the real world, thus enabling researchers to study spatial navigation in controlled virtual tasks, expecting results similar to those in the real world (Aronov & Tank, 2014; Dombeck et al., 2010). This was an important point in our decision process to choose Minecraft for our study.

2.1.2. Siegel & White: the dominant framework

The process of spatial knowledge acquisition is described most famously by Siegel and White in 1975. The authors suggest a framework that became known as the *dominant* framework in the field (Siegel & White, 1975). They propose a three-stage progressive development of spatial knowledge:

- *landmark knowledge*: an initial phase of recognition of specific features of the environment that will later serve as navigational anchors;
- *route knowledge*: a successive phase when landmarks are put into relations between each other, forming what we call a *route*, that is, a sequence of landmarks to encounter when travelling along a path;
- *survey knowledge*: also known as *bird's eye view*, an allocentric representation of the overall environment, which allows for route planning and distance judgments.

Landmarks can be discrete objects, eg, a specific church or restaurant, or more distributed entities, eg, the shape of a building, its colour, or the way it stands out in the space. It is worth noticing that it is possible to navigate also without landmarks, physiologically keeping track of displacement from a base point, as many mammals do, but the error accumulates over time (Epstein et al., 2017). On another note, some populations with strong traditional wayfinding methods, such as the Inuit, can identify more complex features of the landscape and nature as landmarks, such as wind behaviour, tidal cycles, or snowdrift patterns (Aporta & Higgs, 2005).

The impact this framework has had is extensive, and it remains a pillar in the field while also stimulating discussion. Some debate the *staircase hypothesis* of learning order, for example, proposing a model of *continuous learning* in which metric spatial knowledge develops continuously rather than in discrete stages (Montello, 1998). Other later research suggested that ordinal route knowledge must not always precede ordinal survey knowledge, and that different people can develop their spatial knowledge with different levels of exposure, giving more importance to the subjective part and opening the dialogue for a model between the continuous and stage-like proposed ones (Ishikawa & Montello, 2006).

Another discussed dichotomy in the literature is active and passive contributions to spatial learning. Most of spatial acquisition happens incidentally while we move and interact with our environment, as we drive to work, navigate to the supermarket, or walk towards our favourite cafe (Dey et al., 2018). Most importantly, landmark and boundary information appear to be learned without conscious allocation of attention, as object recognition is essentially an automatic process, and landmarks are by definition distinctive features that naturally draw attention. More sophisticated knowledge, such as route and survey knowledge, requires a more deliberate effort into attention allocation (Chrastil & Warren, 2012). Route knowledge develops as we travel through landmarks and gather information about how they are connected in space.

Finally, internal and external factors can impact the process. Internal factors such as age, working memory, and sex, also referred to as *individual differences*, can have a large impact on one's spatial learning performance. Large individual differences in accuracy and in patterns of spatial knowledge development were found in the literature (Ishikawa & Montello, 2006), along with age as a good predictor of landmark positioning accuracy (Muffato et al., 2017).
External factors also play a crucial role in the performance. Starting from the basics, the complexity of an environment can result in different learning efforts; the modality of learning, whether direct navigation or a map, influences the kinds of knowledge acquired, with a focus on either route or survey knowledge (Richardson et al., 1999). Finally, the kind of navigational aid used, as turn-by-turn instructions or map-based, also substantially impacts what is learned incidentally (Dey et al., 2019); this topic will be covered more in detail in Section 2.3.

2.1.3. Reference Frame and location representations

An important concept is that of *reference frame*, which provides *"a way of representing the locations of entities in space"* (Klatzky, 1998); in fact, we cannot point to the location of a given object without using another one as a reference.
From here, we can identify two principal different spatial representations (Dey et al., 2018):

- the *egocentric* representation, where objects' positions are encoded in relation to the observer; this is useful for route acquisition as both environmental cues, eg, landmarks on the way, and possible actions, eg, navigation instructions, can be encoded within the same representation;
- the *allocentric* representation, in which objects' positions are encoded in relation to other objects; this is, on the other hand, useful for route planning, as we can specify the entire environment within a single frame of reference, and include the observer as well (eg, a map).

As an example, when examining the planimetry of a hotel room, we can identify the bed as a few steps away from the observer, following an egocentric representation, or as directly next to the mirror, using an allocentric representation instead. Paper maps, for instance, can encode both types in a single medium, drawing objects at their locations but also adding a "*you are here*" sign to represent the observer's position.

2.2. Spatial Navigation

Spatial navigation is the process encompassing planning a route and moving through space towards a destination (Ishikawa et al., 2008). Although we might not be actively aware of it, as it is a common task, it does require effort and multilevel cognitive processing, from low-level sensory processing (visual perception, motor control) to high-level executive functions (route planning, decision-making). It has hence attracted considerable interest from researchers in the field. The process we call *wayfinding*, which means explicitly choosing a path and following it to reach a destination, involves several stages. First, people need to locate themselves in space, understanding their position and direction; next, they need to plan a route in their head, knowing their goal's location; finally, they execute the route, moving in space following it. All these stages involve using information about the environment, either internally stored knowledge (eg, cognitive maps), external references (eg, paper maps), or a mix of the two.

2.2.1. Navigation Strategy Types

Building on what we previously mentioned in Section 1, let us further investigate the two navigation strategies, spatial memory and stimulus-response, each relying on distinct cognitive processes and brain systems (Dahmani & Bohbot, 2020).

The spatial memory strategy uses cognitive maps in the brain to plan routes and navigate the environment. It makes heavy use of the hippocampus and depends on the person's spatial memory. The stimulus-response strategy, on the other hand, works by associating a stimulus (e.g., seeing a specific building) with an action (e.g., turning left) and relies on the caudate nucleus, a brain region involved in habit learning.

These two strategies are not exclusive, but different situations can favour one over the other: stimulus-response is convenient for frequently travelled routes, as it allows individuals to navigate in "auto-pilot" mode, associating motor responses to environmental stimuli. This strategy is similar to using GPS turn-by-turn directions, which we discuss in the next section.

2.3. Navigation Aids

A *navigation aid* is a device or interface that helps individuals in their spatial navigation tasks. Different types of aids trigger different cognitive mechanisms and thus have different impacts on our spatial memory, wayfinding skills, spatial knowledge acquisition, and so on. Some examples are paper maps, GPS-based interfaces, auditory cues, and *Augmented Reality* (AR).

2.3.1. Paper maps

Paper maps have been around for centuries, representing, in fact, a traditional way to describe space through a physical medium. Due to their static nature, created with a fixed alignment, the knowledge acquired from it is orientation-specific and requires an effort to adapt it to different perspectives (Richardson et al., 1999). It also requires users to align the map representation with the surrounding environment, internally updating their position during navigation, and mentally plan routes, which has a cost in terms of effort but benefits in terms of acquired configurational survey knowledge (Ishikawa et al., 2008). Digital interfaces that rotate with the user's orientation (as in our experimental design) can provide orientation-free representations similar to egocentric navigation.

2.3.2. Digital Interfaces

Digital interfaces are largely outcompeting paper maps (Huston & Hamburger, 2023). With the introduction of *TomTom* in the 1990s, portable navigation devices became commercially available to the general public, making car navigation more accessible to everyone. As technology improved, integrated car systems grew in popularity, and smartphone apps followed (e.g., Google Maps). Given the ubiquity of digital navigation interfaces and devices, this section focuses great effort on this topic.

A classical Ishikawa's study investigated users' wayfinding behaviour and spatial knowledge in real-world walking tasks, comparing paper maps and direct route experience with GPS-based mobile navigation (Ishikawa et al., 2008). Their results suggested poorer performance in both wayfinding (longer travel distance, more stops) and survey knowledge (larger direction error, poorer sketch map drawing) among GPS users. This is, however, the oldest study (2008) among those mentioned, and technology has evolved. First, the experiment was essentially the first time users would use such a system, which would be unlikely nowadays. Furthermore, the device used had a smaller screen size (4 x 5 cm) than the ones available today. Despite these limitations, their findings are still important to understanding how navigational aids affect spatial knowledge acquisition.

A later study by Ishikawa focused on the long-term effects of GPS reliance, questioning whether habitual use deteriorates wayfinding skills (Ishikawa, 2019). After completing a series of questionnaires regarding sense of direction, mental rotation, and frequency of GPS use, participants would take part in a real-world navigation task.
Results showed lower path efficiency and recall accuracy for habitual in-car navigation users when travelling with paper maps. People with stronger mental rotation or sense of direction committed fewer errors and made better direction estimates. The adverse effects may stem from repeatedly following turn-by-turn instructions passively without active decision-making and reduced environmental understanding.

Hejtmánek's eye-tracking study explored how attention to navigational aids, specifically a GPS-like map, influences navigation performance in a virtual environment (Hejtmánek et al., 2018). Participants completed different tasks consisting of a learning phase (going from point A to point B with a guiding line) and a recall phase (going back to point A). Participants needed to follow the trail during the learning phase, but were free to return however they preferred: provided routes were not the most efficient ones, allowing users to find faster ways.
Results aligned with other findings: greater GPS use correlated with less accurate spatial knowledge and longer travelled paths when the aid was removed. Both location naming and placement scores declined with greater map use during learning, indicating impairment across knowledge types. Map use during learning, along with measured spatial abilities, was one of the strongest predictors of performance, suggesting that active environmental engagement during learning would have produced better navigation performance and more precise mental representations.

Several explanations for spatial memory impairment have been proposed, including decreased environmental engagement, reduced active mental effort, and divided attention. Gardony focused on the latter, investigating navigational efficiency and spatial memory in a virtual navigation task and discussing the concept of divided attention (Gardony et al., 2015). They crossed two pairs of conditions: divided versus undivided attention and aided versus unaided, resulting in four tasks. Participants would hear the navigation aid as an auditory direction cue ("in 200m, turn left"), but in the divided attention condition, the aid would be nonsense phrases.
Navigation aids increased efficiency when attention was undivided but impaired performance when divided. Results suggested that aid presence impairs spatial memory; however, when at-

tention was already divided, aid presence did not further impair it. The authors propose that navigational aids divide attention sufficiently to impair spatial memory formation, as sustained attention to the environment is required for effective spatial learning, while navigational aids can bias attention towards themselves.

Interface components can be engineered to understand their impact on navigation. Dey investigated how relative location updates and turn-by-turn instructions (as directional arrows) affect incidental spatial learning, comparing *map-based* and *video-based* interfaces (Dey et al., 2018). Results suggested that relative location updates, also integrated in our interfaces, encouraged active spatial processing, while turn-by-turn directions provided limited benefits. Map-based interfaces resulted in better survey knowledge, while video-based interfaces supported route knowledge, aligning with the literature.

Kapaj explored the influence of landmark visualisation style on mobile maps, comparing realistic 3D representations with abstract visualisations (Kapaj et al., 2024). They analysed how attention and spatial learning changed, hypothesising that 3D landmarks would yield a more natural depiction of the environment and facilitate the association of map features with real-world features. Although realism did not affect overall performance, it helped lower-spatial-ability users with landmark recognition. Participants made longer fixations on the environment with 3D landmarks, suggesting greater engagement. The authors argue that landmarks' visualisation style can modulate gaze behaviour, which in turn can predict some aspects of spatial learning. Therefore, they advocate for adaptive interface designs tailored to users' spatial abilities.

A longitudinal study by Dahmani and Bohbot found that habitual GPS use leads to poorer spatial memory (Dahmani & Bohbot, 2020), though this evidence is based on a small subsample. Continued use removes the need to continuously update our internal location and to pay attention to the environment, as following directions is sufficient. This resembled a stimulus-response navigation strategy (e.g., "at the end of the road, turn right"), which long-term GPS users tend to over-rely on. When asked to navigate without it, they struggled more. In their study, they find that greater GPS use since initial testing correlated with a steeper decline in hippocampal-dependent spatial memory. Importantly, they argue that extensive GPS use leads to memory decline rather than vice versa, as they found that people did not use more GPS due to self-perceived poor navigation skills.

Studies have also explored other modalities beyond vision, yielding interesting results, especially in the context of driving. Drivers using voice directions arrived at the destination faster and with fewer errors than map users (Streeter et al., 1985). This kind of instruction involves complex cognitive processes such as the degree of trust in the speaking voice (Large & Burnett, 2014), and is shown to improve incidental spatial knowledge acquisition during assisted navigation (Wunderlich et al., 2020).

Recent AR research compared wayfinding in 2D electronic maps and smartphone-based AR aids, finding no performance difference between the two groups, but highlighting higher cognitive load and easier map sketching for participants using 2D maps (Dong et al., 2020).

In summary, navigation aids vary considerably in their interface design and cognitive impact, offering different approaches to supporting navigation tasks. For this study, the most relevant information concerns GPS-based interfaces and their effects on cognition, specifically the strong trend toward spatial memory impairment associated with habitual use. Reliance on turn-by-turn navigation discourages active spatial processing, impairs the development and accuracy of cognitive maps, and negatively affects navigation performance in general. We aim to mitigate these effects with our experimental design exploring a landmark-focused interface alternative.

2.3.3. Impact of individual differences

Spatial navigation performance and spatial knowledge acquisition are influenced by many individual characteristics, which have been extensively studied across the literature. It is crucial to understand these differences to interpret navigation research findings and design effective navigation aids.

Working memory capacity is the ability to maintain information in memory and plays a significant role in navigation performance (Bushinski & Redick, 2025). Individuals with higher working memory capacity showed greater resilience to divided attention when using navigation aids, maintaining better spatial knowledge acquisition even when their attention is split between the interface and the environment (Gardony et al., 2015).

Self-reported spatial abilities are a consistent predictor of performance across a range of spatial learning tasks. As an example, individuals with lower self-reported spatial abilities get a larger benefit from realistic landmark representations, suggesting that interface design can partially compensate for differences in spatial aptitude (Kapaj et al., 2024). This relationship between self-assessment and performance seems to be bidirectional: participants who scored poorly on neurocognitive tests assessing spatial skills also reported lower confidence in their spatial abilities and compensated by fixating more frequently on navigation interfaces rather than looking at the environment (Hejtmánek et al., 2018).

Sense of direction represents another critical point of spatial ability. It is an important correlate of spatial orientation and learning: people with a good sense of direction do not perceive the environment egocentrically, and hence tend to favour survey strategies, while the opposite applies for users with a poorer sense of direction (Ishikawa, 2019). Importantly, spatial aptitudes also correlate with the choice of navigation tools, with frequent users of pedestrian GPS-like systems tending to have a poorer sense of direction and mental rotation abilities, though the causal direction of this relationship is still under debate.

Demographic factors have also been investigated. For example, age has an impact on how people interact with navigation aids. Older adults demonstrate different gaze patterns compared to younger adults, tending to spend less time fixating on navigation systems and more time attending to the physical environment (Dey et al., 2018). This shift in attention allocation may reflect either compensatory strategies or differences in technology familiarity.

Gender differences have also been extensively studied in the context of spatial navigation, with males consistently showing advantages in navigation efficiency and performance (Munion et al., 2019). These differences appear to be mediated by different wayfinding strategies: women tend to prefer landmark-based route strategies, while men more often use orientation strategies based on maintaining a sense of position relative to environmental reference points; women also report higher spatial anxiety related to wayfinding (Lawton, 1994).

Prior experience with video games also has an impact on the performance in virtual navigation tasks. Video game players, particularly those with extensive experience in three-dimensional game environments, demonstrate enhanced spatial cognition skills that transfer to navigation tasks (Yavuz et al., 2024). This advantage probably comes from familiarity within virtual spaces.

Finally, although not specifically related to individual difference, landmark saliency, which consists of how distinctive or noticeable a landmark is, also affects recognition and memory, with visually and cognitively salient landmarks improving navigation and spatial memory (Dommes et al., 2024).

2.4. Transferability between Virtual Environments and the Real World

The concept of *transferability* refers to the extent to which spatial abilities and knowledge acquired in virtual environments transfer to real-world navigation. Researchers have sought to understand whether spatial knowledge transfer occurs between real-world and virtual environments, testing virtual and real versions of the same environment (Clemenson et al., 2020), or, more generally, whether the exact cognitive mechanism underlies both conditions.

Among the aforementioned papers, some would employ real-world navigation tasks, while others would make use of virtual experiences. Many tools are available to build virtual environments, depending on the age of the paper, mostly related to video games. Known mentions are the DOOM engine, a low-graphics engine made famous by the related game and employed by (Richardson et al., 1999), and Unity 3D and Unreal Engine, used by multiple researchers, a widely recognised choice for building 3D environments and video games, also by the AAA industry.

Deploying a task within a virtual environment, as in spatial navigation, offers several benefits over a real-world experiment. First, it allows for a more granular control over the setting of the task, and not to have to worry about external factors such as weather, bypassers, or noise; there is no limitation by the topological structure of an environment, as one can build one's own from scratch according to their needs and choices. Second, it generally takes less time, and it becomes easier to pilot and make small changes on the go.

There are, however, some drawbacks to this choice. A valid concern is about realism and fidelity (Kelly & Gibson, 2007). While low-detail environments can be appropriate for tasks requiring configurational information (e.g., pointing estimation), this configuration was shown to be detrimental for wayfinding and the use of spatial representation (e.g., sketch maps) (Wallet et al., 2011). Another issue addressed by Wallet is active learning: they find that physically engaging with the virtual environment (in their case, having the participant move with a joystick) enhances the transfer of spatial knowledge between the virtual and real environments.

Another known limitation is information mismatch. Inducing an egocentric update requires not only visual information available in the VE but also vestibular information, which is missing when playing on a screen (Richardson et al., 1999). Wrapping up, research suggests that transfer exists but is incomplete (Ahmadpoor & Shahab, 2019), and that patterns observed in VE are likely to transfer to the real world (Hejtmánek et al., 2018).

2.5. The case of Minecraft

The popularity of Minecraft, first published in 2011 by the Swedish indie game studio Mojang, is beyond discussion among players. With over 350 million copies sold, it is the most-sold game in history (AB, 2025). It became a cultural phenomenon, with movies being made and a continuously growing fanbase spanning generations. Minecraft is an open-ended, 3D, procedurally generated, cube-based, virtually infinite game, allowing each game to be set in a different environment. Moreover, simulated physics and simple movement controls make it intuitive and easy to pick up for new players. The cube-based setting allows a map to be modelled according to research needs, not only by building custom-crafted spaces but also by utilising Minecraft's built-in natural biomes (e.g., forest, plains, ocean) and world types (e.g., super-flat, with caves).

Research has not yet extensively explored this game's capabilities. A recent study analysed spatial reasoning skills collecting gaze data and user actions (Andrus et al., 2020), while Stark focused on hippocampal-associated memory questioning whether a small amount of daily gameplay could reduce age-related cognitive decline in middle-aged adults (Stark et al., 2021). A more recent study explored the role of sleep on spatial memory and navigation using Minecraft, observing a facilitation of the former but no specific impact on the latter (Simon et al., 2022). Finally, a pilot study proposes a Minecraft mod as a system for early detection of Alzheimer's disease, by analysing players' spatial navigation patterns (Ito et al., 2022).

For our research purposes, the aforementioned advantages of Minecraft make it an ideal testbed for investigating how interface design affects spatial knowledge acquisition, in a context that offers extensive experimental control and provides potential for real-world relevance.

3. Methods

This chapter provides a detailed description of the methods employed throughout the study. We analyse the participants' sample and our laboratory setup at both the hardware and software levels. We then provide in-depth descriptions of the independent and dependent variables, the experimental procedure, and the data analysis techniques.

3.1. Participants

A total of 16 subjects participated in the experiment, recruited randomly from university students and our acquaintances. No specific criteria were applied in the selection process, apart from the ability to move in a virtual world in a video game; this was generally tested by directly asking, and, in the case of no experience with video games at all, with a quick Minecraft test unrelated to the tasks. The subjects were between 20 and 32 years old, from diverse educational backgrounds and countries; 7 of 16 were females, while the remaining 9 were males, providing a somewhat balanced setup. No compensation was provided for participation.

3.2. Setup

The study was conducted at the University of Copenhagen DigiLabs - Observe Room, which is equipped with an eye-tracking device, Tobii Pro Spectrum[1] and Tobii Pro Lab software for experiment design, gaze data analysis, and metrics export. This eye-tracking device can record eye movements at 1200Hz and allows participants to wear corrective glasses and interact freely with a computer. A calibration phase preceded each task. Participants were first required to sit at an appropriate distance from the device so their eyes could be correctly tracked, and then they would see a small dot moving around the screen that they needed to follow with their eyes. Participants were asked to recalibrate if the detected data loss exceeded 5%, the same threshold as (Hejtmánek et al., 2018). Participants would play on the built-in Spectrum screen, 23.8″ 1920x1080 resolution, interacting with a generic keyboard and mouse.

3.2.1. Minecraft

On the machine, a modded Minecraft 1.20.1 version with Forge 47.4.0 was installed and loaded through the CurseForge launcher. This specific version was chosen because of its compatibility with Xaero's mods.

As the concept of *minimap* is not native to Vanilla Minecraft, three mods from the same ecosystem were used for the experiment: Xaero's Minimap, Xaero's WorldMap, and a custom version of Xaero Plus. The mods implement a large set of map-related features in Minecraft, most notably landmarks and map-drawing utilities. Being Xaero Plus code open-source[2] allowed for further customisation to better suit the experiment's needs. In particular, we implemented the following functions:

- a command `/paths <id>` to load a `path<id>.json` file containing a sequence of coordinates, and then draw a coloured line between them on the minimap to mimic GPS systems' navigation routes;
- a command `/lobby` to teleport the player back to the lobby, for convenience, as tasks were quite spread around the map; the lobby would contain shortcuts to teleport to each task starting point;
- two commands `/tbt_on` and `/tbt_off` to toggle navigation instructions on-screen; the interface is detailed in Section 3.3.1;

[1]Details and specifications on the official website

[2]Original code available on the official repository, custom fork available on my personal GitHub

- a command `/capture_orientation` to print to chat the yaw value of player orientation, that could be triggered by pressing `0`.

The commands to navigate to the lobby and those to initialise paths and interfaces would only be run by us experimenters before a task started. In case of accidental press of the `0` key, the event would be manually noted down and later removed from the log parsing.

The Minecraft world played is a custom map called *Vertoak City* made by user *Fish95* and available on Minecraft Forum. The map depicts an imaginary island with cities, smaller villages, and mansions. It thus provides many of the structures and topological complexity we were looking for in a virtual city. We further modified the map to better fit the experiment, for example, by adding blocks to make landmarks more recognisable (e.g., adding chairs and tables outside the pub, or building a cross on top of a church) and by building totem structures at the start and end of each task.

The game was always played in creative mode, with auto-jump enabled to simplify the experience, so players would not need to worry about vertical navigation or introduce an additional button to press, namely, the space bar. The weather and daylight cycle are enabled by default in Minecraft: they emulate the regular alternation of day and night, and introduce weather randomness by adding rain and storms. To ensure a consistent experience for players, these two cycles were disabled, forcing the game to daylight and clear skies.

The performance tracking logic was implemented in-game using *command blocks*. This specific Minecraft block[3] allows command execution when triggered by a Minecraft circuit, for example, via a button. It also uses chat text events, which are logged with a timestamp and stored in the game directory.
A *starting point totem* and *end goal totem* structures were built and placed at the extremes of each task. They are similar in functionality but differ slightly in location and consist of three wooden sections, each with a button that triggers a command block and a sign above it describing the button's functionality.

Figure 1: Starting point totem in task 1. We can see the three buttons, each with a descriptive sign above it.

[3]More information about command blocks on the official Minecraft wiki

Specifically, as visible in Figure 1, each totem would have:

- a button to *start* a task phase: at the starting point, it would start phase one, while at the end goal, it would start phase two;
- a button to *end* a task phase: at the end goal, it would end phase one, while at the starting point, it would end phase two;
- a button for the *pointing task*, telling the user to look towards the point they came from and press `O` to capture the decided direction.

At the end goal spot, the sign above the pointing task button would also remind the user to press `P` on the keyboard to hide the minimap before continuing; this was initially planned to be automated without user interaction, but the implementation did not work out. Users would be thoroughly instructed on the procedure, which is described in Section 3.4.

Each button would trigger a command block that writes a description of the selected action to the chat, e.g., *"starting task 2, phase 2"*. Although quite simple, it serves two functions. First, provide the user with visual feedback on the task's state, letting them know when a new phase has started and that they need to act accordingly. Second, and more importantly, allow detailed performance tracking by simply looking at the game logs. Here is a snippet from what the logs look like:

```
[17:13:29] [Server thread/INFO]: Saving and pausing game...
[17:13:34] [Render thread/INFO]: [System] [CHAT] Teleported to lobby.
[17:13:45] [Render thread/INFO]: [System] [CHAT] Teleported to lobby.
[17:13:45] [Server thread/INFO]: [Bragg13: Teleported Bragg13 to -563.500000,
71.000000, 867.500000]
[17:13:45] [Render thread/INFO]: [System] [CHAT] Teleported Bragg13 to
-563.500000, 71.000000, 867.500000
[17:13:51] [Render thread/INFO]: [System] [CHAT] Path mode set to LF
[17:17:32] [Render thread/INFO]: [System] [CHAT] [@] start task 3 - part one
[17:18:47] [Render thread/INFO]: [System] [CHAT] [@] end task 3 - part one
[17:18:51] [Render thread/INFO]: [System] [CHAT] [@] orientate towards your
starting point, then press O to capture direction
[17:18:55] [Render thread/INFO]: [System] [CHAT] angle=141.7
[17:19:00] [Render thread/INFO]: [System] [CHAT] [@] start task 3 - part two
[17:19:49] [Render thread/INFO]: [System] [CHAT] [@] end task 3 - part two
[17:19:51] [Render thread/INFO]: [System] [CHAT] [@] orientate towards the goal,
then press O to capture direction
[17:19:57] [Render thread/INFO]: [System] [CHAT] angle=58.9
[17:20:16] [Render thread/INFO]: [System] [CHAT] Teleported to lobby.
```

Listing 1: Snippet from Minecraft log file. The first line is related to game loading, while the rest presents events relevant for our analysis, such as *start task 3* and *angle=141.7*.

The first pair of square brackets in Listing 1 indicates the time of the event, while the text at the end of the line reports the message written to chat. The other brackets contain information irrelevant to the study. From this point, it is pretty straightforward to calculate the travel time for each phase, the time spent deciding the angle (*pointing* or *decision* time), and take note of the pointing angle. Furthermore, it provides a uniform, native measurement tool.

In the event of a user mis-click, we would note the event down and ignore it during log parsing. A testing task was also created to familiarise the user with Minecraft and its navigation interface. Nine participants out of 16 reported never having played Minecraft, and one had never played any video game. It was thus crucial to have a testing Minecraft world where people could try the controls, adjust mouse sensitivity if needed, and become familiar with in-game movement. There was never a problem of a player being completely unable to play the game that would require their disqualification from the study.

3.3. Experimental Design and Variables

In this section, more details on the experimental design are provided, building on the brief description provided in Section 1.3. As mentioned earlier, the current study investigates the impact of two different navigational interfaces across four tasks of different difficulty. We describe the interfaces and tasks, along with the initial pilot study design. Then, we give an overview of the dependent variables, the task-interface assignation method, and the data analysis process.

3.3.1. Interfaces

Two different interfaces were designed for the experiment, namely *Landmark Focused (LF)* and *Turn-by-Turn (TBT)*. Both feature a square minimap in the top-right corner of the screen, showing the map from above, similarly to how a satellite layer in Google Maps works; all buildings, roads, and natural features are thus visible in a top-down fashion. On top of this, we can see *landmarks*. The minimap is not fixed to a specific orientation; instead, it rotates with the player's head movement, keeping the player's arrow on the minimap facing north, as one would expect from a navigation application.

Two principal characteristics distinguish TBT from LF. First, a coloured path line is drawn on the minimap. It highlights the route from the starting point to the end goal, allowing users to follow it to reach their destination. The line is rendered at the beginning of the path and remains unchanged throughout, unlike modern navigation apps, where the blue line disappears as you follow it. The second one is a rectangular box appearing in the top-right corner of the screen, just above the minimap, containing a short string indicating where to go next (e.g., "turn ➡️, then ⬅️"). These *navigation cues* would pop up at specific coordinates along the route, spaced roughly equally. The number of alerts was between four and six, depending on the task. These two features were designed to mimic a familiar GPS interface, such as the one in Google Maps, for in-car or pedestrian navigation.

Landmark-focused (LF) interface — Turn-by-turn (TBT) interface

Figure 2: In-game screenshots of the two interfaces taken on the testing task.

Landmarks are visible as names on the minimap and are designed to stand out in the world and be recognisable; this mimics real-world scenarios, e.g., a pin on Google Maps labelled "*Restaurant*" would be located at the same spot as the actual restaurant. A fundamental point participants were informed of is the strategic placement of the landmarks on the map. Specifically, the order of the landmarks is designed to create a route users can follow to reach the task goal; this allows players to follow the landmarks one after another to reach their destination without additional aid, which is precisely the behaviour encouraged by the LF interface.

As part of the effort to align the interfaces with familiar ones and make the information intuitively accessible, landmark names are displayed on a coloured background. They are divided into five pseudo-categories:

- gold is for infrastructures (eg, a bridge, a harbour)
- blue is for shops (eg, plant shop, fish shop)
- green is nature-related (eg, park)
- purple is for the task's goal
- red is for other, more generic places (eg, hotel, bar)

The colour coding was not explicitly provided to the player because it was not strictly necessary to accomplish the task. Furthermore, it would require more information to remember and deviate from the normal experience of using a GPS application, where landmark colours are intuitively inferred by the user and not strictly required for travel. It served only as a means to further stimulate familiarity with the interface.

3.3.2. Tasks

Four tasks were designed to maximise the amount of collectable data, while keeping a reasonable time length, so as not to excessively tire participants. A single phase of a task, excluding totem interactions, takes, on average, 73 seconds to complete, though this can vary depending on task difficulty. Tasks are located in four areas of the map to avoid contamination and to provide a unique landscape that reduces confusion between them.
An overview is available in Figure 5.

Figure 5: A top-down view of each task map, with the landmarks list shown on the left. The red dot represents the starting point. Note how tasks 1 and 2 develop vertically, while tasks 3 and 4 develop horizontally. This is the interface shown to participants after completing a navigation task: they are required to draw their route on the map and drag each landmark to its correct position.

Tasks are described as follows:

- *Sea Village* (task 1) is set on the shore of a large body of water (to the east), mountains surround it, and for a large part, the road follows railroad tracks. Furthermore, it is the only task in which the starting point is clearly visible from the end goal.
- *Lake Town* (task 2) is set between two small bodies of water and is surrounded by a low jungle. It has wide roads and is densely filled with tall buildings.
- *Desert Stronghold* (task 3) is set in the desert, with a big set of stairs in the beginning, and a flat route unfolding in a regular grid layout. It is surrounded by sand and portrays a high colour homogeneity, as the buildings are also made of sand blocks.
- *City Suburbs* (task 4) is set on one end of a city and borders a body of water. Its geometry is more complex, as elevation changes and most turns are not sharp. It also has less prominent landmarks, such as White Tree or Sea BnB.

More detailed information is available in Table 1.

Task	Name	Length (blocks)	Number of turns	Number of landmarks	Landmarks
1	Sea Village	200	6	6	Pub, Fish Shop, Dorm, Harbour, and Goal
2	Lake Town	260	8	6	Veterinary, Church, Hotel, Plant shop, Library, and Goal
3	Desert Stronghold	190	4	4	Oak temple, Sun temple, Restaurant, and Goal
4	City Suburbs	200	6	6	Sea Bnb, Park, White Tree, Art Gallery, Fountain, and Goal

Table 1: Each task is indicated along with its name, difficulty settings, and landmarks.

Difficulty was originally calculated in terms of three factors. The first we consider is *route length*, manually measured using in-game map tools provided by Xaero's mods, expressed in blocks, as they are the basic unit of measure in Minecraft. Second, we consider the *number of turns*, that is, the decision points at which the direction changes and users need to turn. Third, we consider the *number of landmarks* including the goals. Higher values, indicating more elements to remember and higher route complexity, were considered more challenging. Each difficulty factor was summed and divided by the maximum value across tasks to obtain a value between 0 and 1.

Let us take task 1 as an example:

- task length of 200 blocks divided by 260, the maximum task length (found in task 2), gives a score of 0.76;
- number of turns of 6 divided by 8 (found in task 2) gives a score of 0.75;
- number of landmarks is 6, divided by 6 (found in Tasks 1, 2, and 4) gives a score of 1.0;

Summing up $0.75 + 0.76 + 1.0 = 2.51$, divided by the number of factors results in a difficulty score of $2.51 \div 3 = 0.84$.

The resulting difficulty scale shows task 3 as the easiest (0.63), followed by tasks 1 and 4 (0.84), and finally task 2 as the most difficult (1.0). Other difficulty scales were calculated post-hoc, resulting in similar scales to the original design, and are discussed in Section 5.1.1

The testing task was designed to familiarise players with the interfaces and to use landmarks to navigate. It is a simple task with a length of 130 blocks, four turns, and five landmarks, namely Tower, BnB, Sword Shop, Vegetable Garden, and Goal. It is set in a Minecraft village generated structure, with a plains biome serving as a simple landscape to get used to the game. Its difficulty score is 0.61.

3.3.3. Pilot

Our initial pilot design consisted of three tasks of varying difficulty and considerable length: the number of blocks ranged from 350 to 420, while the current longest task consists of 260 blocks. The number of landmarks ranged from 12 to 14, whereas the current maximum is 6. We changed how landmarks are displayed on the minimap. Only an emoji would be displayed instead of the landmark's name, mimicking the system of symbols in *points of interest* in modern GPS applications. The directional instructions in the turn-by-turn interface would appear in the centre bottom of the screen on the *action bar* using Minecraft APIs[4]. The landscape setting of tasks was also different: they all took place in a big city, whereas current tasks are set in smaller towns, far apart from each other, each with a specific landscape surrounding it.

We conducted a preliminary pilot test with a single participant to evaluate the task design, assess perceived difficulty and overall feasibility, and test the data collection process. It was impossible to complete any of the tasks in phase 2 in either interface; in fact, the landmark-focused interface could not even be used for phase 1.

Several reasons were involved. A very long route proved to be too difficult to remember, and the same goes for the landmarks. A number of six seemed more reasonable as it more closely aligns with the famous working memory capacity of 7 (Miller, 1956).
Landmarks were also not very noticeable in most cases, as reported by the pilot participant, hence making recognition challenging even before remembering.
The minimap visualisation added another layer of challenge. As only emojis were displayed, participants needed to not only notice the landmark on the minimap and associate it with a structure in the world, but also give the emoji symbol a meaningful semantic interpretation, for example, understanding that the emoji "🛏️" refers to a hotel. Additionally, Minecraft is a cubic world, so associating a structure with its actual meaning can require intuition or experience with the game.

These visualisation issues are the reason why a strong focus was placed during the design of the current tasks to make the landmarks as noticeable as possible. This process not only involved choosing more detailed buildings and heavily decorating them, but also moving the task locations. The city centre, where tasks were initially set, is a chaotic environment with few distinctive structures. We aimed to reduce confusion about routes and landmarks by choosing a distinctive environment for each task and seeking less distracting surroundings. The directional instructions in TBT were moved to the top right of the screen to have them close with the minimap, as it would be the case for a navigation application; this was implemented as *advancements*

[4]Detailed information on Minecraft Wiki - Action Bar

using Minecraft APIs[5].
Finally, the tasks are now four, with each participant playing twice with each interface, as described in Section 3.3.5.

3.3.4. Dependent Variables

The dependent variables chosen for this study are commonly found in the literature for this kind of experiment (Dey et al., 2018; Dong et al., 2021; Gardony et al., 2015; Hejtmanek et al., 2020). They measure spatial knowledge (landmark placement, orientation angle error and decision time), spatial navigation performance (time to complete a task), and individual skills differences (SBSOD, PTSOT, Videogames and Minecraft experience). Further references are provided for each variable.

Navigation time Also called *travel time*, it is defined as the time it takes a player to travel from the starting point to the end goal of a task, in seconds. Used as the primary metric for navigation performance, a lower time generally means higher confidence in navigating the environment. Navigation time is used as a performance tracker across the literature (Dey et al., 2018; Dong et al., 2020; 2021; Hejtmánek et al., 2018; Richardson et al., 1999).
Having the travel time for each phase allows us to define an additional variable, *recall extra time*, as follows: $t_{\text{phase2}} - t_{\text{phase1}}$. A low recall extra time indicates that returning to the starting point did not require much more effort without a navigation aid; a negative recall extra time means that a player was faster to return without a navigation aid than to arrive at the destination in the first phase.

Landmark placement error Also called *map sketching*. After each task, participants were presented with a blank top-view map of the environment they had just navigated (the same as shown in Figure 5) and asked to draw the route they had taken. A list of task landmarks was also provided, and they were asked to drag them to their correct positions. The map presented the task starting point as a red dot for facilitation. Accuracy was assessed by measuring the pixel distance between the correct coordinates and the user-selected position. The route was excluded from the calculation due to time constraints.
The idea is similar to what is done in literature (Dey et al., 2018; Gardony et al., 2015; Hejtmánek et al., 2018; Thorndyke & Hayes-Roth, 1982).

Pointing angle error and speed A pointing task was performed at the end of each phase. Once participants reached the end goal (or the starting point on the way back) and completed the first phase (or the second), they would be asked to aim the crosshair at their point of origin. Upon pressing the key `0`, the direction faced by the player, represented by its orientation's yaw value, would be captured and printed to chat. This value is later subtracted from the correct angles, yielding an absolute angle error in degrees, used to measure survey knowledge; smaller angle errors indicate a more accurate knowledge. The time the participant would take to decide on the angle would also be logged and defined as pointing speed or decision time.
The procedure follows the literature (Dong et al., 2021; Hejtmanek et al., 2020; Hejtmánek et al., 2018)

Santa Barbara Sense of Direction Participants completed the Santa Barbara Sense of Direction (SBSOD), developed by (Hegarty et al., 2002), which consists of 15 statements rated on a 1-7 Likert scale, ranging from "*strongly disagree*" to "*strongly agree*". It contains questions about navigation skills and sense of direction, for example, "*I am very good at judging distances*" or "*I have trouble understanding directions*", and provides a measure between 1 (poor sense of direction) and 7 (good sense of direction); high scores are associated with better real-world navigation and spa-

[5]Detailed information on Minecraft Wiki - Advancements

tial knowledge performance. Questionnaires were evaluated according to the original guidelines: reversing the scores of a subset of questions, summing them, and dividing by 15.

Perspective Taking/Spatial Orientation Test Subjects completed the Perspective Taking/Spatial Orientation Test (PTSOT) to measure mental rotation and perspective-taking abilities using the average angle error in degrees. A lower average error is associated with higher precision and, thus, better mental rotation ability. It consists of twelve trials where participants are shown doodles of various objects (eg. a flow, a car, a traffic light) arranged in a circle; for each trial, they are shown an instruction like *"Imagine you are standing at <object1> and facing <object2>; now point to <object3>"*.
Developed initially by (Hegarty & Waller, 2004) and (Kozhevnikov & Hegarty, 2001), a computerised version introduced by (Friedman et al., 2020) renders the measurement easier, providing an interactive Python application[6]. No questionnaires were discarded, as all of them showed a performance above chance ($< 90°$).

Video games and Minecraft experience A discrete measure of a person's familiarity with Minecraft and video games in general. It was manually administered by directly asking the participants how they would rate their experience of Minecraft and video games, separately, on a scale between "*I never played*", "*I seldom play*", "*I play often*", and assigning a discrete value between 0 and 2 accordingly.

Number and duration of fixations (gaze) The amount of fixations on a given *area of interest (AOI)*, which are different sections of the screen we are interested in comparing. Since the absolute number of fixations can be different depending on the participant and the task, we calculated a ratio of fixations for a given AOI, task, and interface, dividing it by the total. Fixation duration is the average time, in milliseconds, spent on a fixation.

[6]Code is available on GitHub

3.3.5. Task assignation

We used a mixed within-subjects design, in which each player played two of the four tasks with one interface and the remaining two with the other. This structure allowed us to collect the same amount of data per interface and task; furthermore, players' subjective skills are equally distributed across the tasks and interfaces they played. The assignments of tasks and interface to each player are visible in Table 2.

Participant	1st position	2nd position	3rd position	4th position
1	1	2	3	4
2	4	1	2	3
3	3	4	1	2
4	2	3	4	1
5	1	2	3	4
6	4	1	2	3
7	3	4	1	2
8	2	1	3	4
9	3	4	1	2
10	4	2	1	3
11	1	4	2	3
12	2	3	4	1
13	3	1	4	2
14	4	1	2	3
15	1	2	3	4
16	2	3	4	1

Table 2: Task-Interface assignments. TBT in *blue*, LF in *orange*. For each participant, we can see which succession of tasks they played and in which interface.

3.4. Procedure

This chapter goes through the unfolding of an experiment, focusing on the steps that make up a task.

3.4.1. Preliminary part

Upon arrival at the laboratory, the participant is first introduced to the experiment. They are informed that it is about spatial navigation, given an explanation, and told they will play Minecraft to perform some navigation tasks. Then, they sign a consent form to use their eye-tracking data, are administered the SBSOD questionnaire, and then complete the PTSOT test digitally. This part takes between five and ten minutes.

Then, the user is taken to the eye-tracking station and presented with the testing task. Before starting to play, they are given more in-depth information about the experiment, e.g., the task structure, the differences between the two phases, and what they will do. They are told about the importance of landmarks and how to follow them to reach the end goal, along with additional information, such as that their performance is timed, and instructions on how to use the start and end totems. They are encouraged to start playing to explore the key controls and the interface; when ready, participants go through the testing task, assisted by the experimenter.

Once done, the real tasks can begin. The experiment lasts between 45 minutes and one hour.

3.4.2. Task unfolding

Before each task, we would manually move the player to the task's starting point in Minecraft, turn on the minimap, enable the correct set of landmarks, and turn on the chosen interface, as specified in Table 2. Then, the eye-tracking calibration can start, and once done, the participant opens the game and finds themselves at the *starting point*.

In any given session, a player would proceed like this:

```
1. Click on the "start task 1, phase 1" button
2. Navigate towards the end goal
3. Click on the "end task 1, phase 1" button
4. Press P on the keyboard to hide the minimap
5. Click on the "in-between tasks" button
6. Orient the crosshair towards the starting point, and press O on the keyboard to
confirm
7. Click on the "start task 1, phase 2" button
8. Navigate back towards the starting point
9. Click on the "end task 1, phase 2" button
10. Click on the "after task" button
11. Orient the crosshair towards the end goal, and click O on the keyboard to
confirm
```

Since we were monitoring the correct unfolding of the task, there was no case of participants navigating the recall phase with their minimap visible or forgetting to click any button.

3.5. Data preprocessing and analysis

The data processing and analysis pipeline was implemented entirely in Python, using a set of the most well-known libraries for data science, such as *pandas* and *numpy* for dataframe manipulation, *matplotlib* and *seaborn* for plotting, and *statsmodels* and *scipy* for statistical investigation. Gaze data was also analysed with *Rstudio*.

For each participant, we put together, as CSV files:

- *performance* data containing angle error, angle time, and completion time for each task and phase. These were calculated and noted starting from Minecraft logs, as shown previously in Section 3.2.1.
- *sketch* data listing all landmarks, divided by task, with the placement error in pixels next to each landmark name. This would be manually measured using the *ruler* tool in *Photopea*, an online image editing application that allows for overlapping a participant's sketch map with the correct landmark locations.

Figure 6: View when calculating the landmark placement distances in Photopea. We can see the starting point (red dot), the landmarks placed by the user (coloured rectangles on the map), and the correct locations (green dots). We can see a thinner line between “Fish Shop” and a green dot on the bottom, which is the ruler tool used to calculate the error in pixels.

A *general* CSV file contains information about the order and assignment of tasks and interfaces for each player, along with their SBSOD and PTSOT scores, and Minecraft and Videogames familiarity scores.
Finally, a CSV file contains the *eye-tracking* exported metrics from Tobii Pro Lab. It consists of the amount of fixation, fixation duration, and other data about glances and visits, divided per participant, task, phase, and *Area of Interest (AOI)*, which could be *minimap* (the interface), *direction* (the turn-by-turn instructions above the minimap), and *env* (the environment, the rest of the screen).

A first phase of the analysis consisted of loading, cleaning, and aggregating the data into dataframes. Names were replaced with numbers to avoid personally connected data.

For reference, a sample of the performance dataframe can be seen in Figure 7.

			time	angle_err	angle_time	iface	phase1_time	phase2_extra	sketch_err
name	task	phase							
1	1	1	80	6.0	9	tbt	80.0	4.0	54.667
		2	84	0.0	9	tbt	80.0	4.0	54.667
	2	1	83	32.0	4	lf	83.0	-8.0	258.500
		2	75	35.0	8	lf	83.0	-8.0	258.500

Figure 7: First rows of `df_performance` dataframe, indexed by name, task, and phase.

where

- each row is indexed by the participant’s name, the task, and the phase;
- time measures, in seconds, include `time` (the total time for that phase), `phase1_time` (the time for phase 1), and `phase2_extra` (the additional time spent on phase 2 compared to phase 1);
- pointing measures include `angle_err` (the error between the correct angle and the picked one by the player, in degrees) and `angle_time` (the time spent picking the angle, in seconds);

- `iface`, the interface played;
- the sketch error from the drawing task, as `sketch_err`, in pixels

A detailed overview of other dataframes is omitted from the main manuscript; please refer to Section A.1. The complete analysis files are available on GitHub.

Thereafter, we conducted preliminary analyses of outliers and task difficulty, then focused on performance and eye-tracking variables.

3.5.1. Outliers

With such a small sample ($N = 16$), outlying values can strongly affect the results and undermine significance. We wanted to understand how many outliers are in our data, whether they are specific to an interface, a task, or a dependent variable, and to identify the reason something went differently in that experiment. We calculated the z-score for each column of the performance dataframe to make it easier to compare and identify outliers. The threshold for a value to be considered outlying is 3 standard deviations, a standard rule of thumb.

Five of 128 trials exceeded this threshold. We kept them in our data; further details are discussed in Section 5.3.9.

3.5.2. Analysis techniques

As mentioned in Section 3.3.5, our experimental design is within-subjects for both interfaces and task difficulty. Each participant experiences LF and TBT interfaces the same number of times, and plays all tasks, although in different orders and assignments. This way, each subject effectively acts as their own control: stable variability in navigational skills affects every trial across tasks and interfaces, rather than confounding group differences. While we cannot ensure that all conditions are equally affected, as a specific interface or task may be more prone to some effects, this is still fairer than randomly splitting participants into two groups based on interfaces. Individual differences are measured to some extent with preliminary questionnaires (e.g., SBSOD), but are not the primary focus of the study. This choice is important for a spatial navigation study, where these differences can have a substantial impact on performance.

Performance For our primary analysis, we used *Linear Mixed-Effects Models (LMMs)*. These statistical models can include both fixed effects (e.g., interface, task) and random effects (e.g., individual differences), and are particularly useful in experimental designs with hierarchically structured data. They do not assume independent observations as traditional ANOVA regression models do, and can explicitly model variability at different levels of the hierarchy. This approach is powerful for our design, as we can capture individual differences with random intercepts for each participant, while investigating the effects of task difficulty, interface type, and their interactions.

The model selection process was iterative. We started with a simple model, serving as our baseline, namely one that treats the interface as fixed effects and includes a random intercept. Then, we gradually added fixed and random effects, testing whether each improvement in fit using the *Likelihood Ratio Test (LRT)*. We stopped and selected the best model when adding more complexity was not justified by the data.

The models are reported for clarity in Table 3:

Name	Fixed Effects	Random Effects	Question
M1 (baseline)	Interface	Random Intercept	Does interface matter at all?
M2	Interface + Task	Random Intercept	Does task difficulty matter?
M3	Interface × Task	Random Intercept	Does interface effect depend on task difficulty?
M4	Interface × Task	Random Intercept + Random Slope (Interface)	Does interface effect vary between participants?

Table 3: Models tested during the iterative selection process.

Each model includes a random intercept to account for participants' variability.

After finding the best model for each variable, we ran Wald's joint test to detect the main effects of interface, task, and interface-task interactions. In the single case where the tests would show a significant interaction, we calculated simple effects to understand interface effects on each task. Interaction terms are provided by the LMM fit and used to draw plots in Section 4.2.

An alternative approach we briefly pursued was a more general, shallow comparison between interfaces. It consisted of "squashing" the tasks together into a single one, reducing the original design's complexity. These results are omitted from the main manuscript and available in Section A.2.

Gaze Eye-tracking data were not normally distributed, thus we could not proceed in the same way as we did with performance variables. We employed *Generalised Linear Mixed Models (GLMM)* with different functions. We used the Poisson distribution, designed for discrete count data, for fixation counts, and the Gamma distribution with log link, for positive continuous data, for fixation duration.

We compared different models using the Likelihood Ratio Test.
For *number of fixations*, we chose M2 as the main effects of interface, task, and AOI, with 2-way interactions, defined as `m2: n_fixations ~ (iface + task + aoi)^2 + (1 | participant)`.
For *fixation duration*, we selected M1 as the main effects of interface, task, and AOI, which was defined as `m1: avg_dur_fixations ~ iface + task + aoi + (1 | name)`, as more complex models did not converge.

Complete results are available in Section A.5. We ran Wald's tests to detect main effects of interface, task, and AOI, along with their interactions, when possible.

4. Results

This section describes the obtained results and illustrates the plots. We start by presenting our analyses regarding task difficulty, then proceed to examine performance metrics, our main investigation. Finally, we show some results regarding gaze, although we did not perform an in-depth analysis.

The p-value threshold level chosen is 0.05.

4.1. Regarding Task Difficulty

We visualised landmark placement accuracy, as we are interested in seeing how well landmarks are remembered across tasks. Plot in Figure 8 shows landmarks placement error sorted from least to most remembered. The values are z-scored.

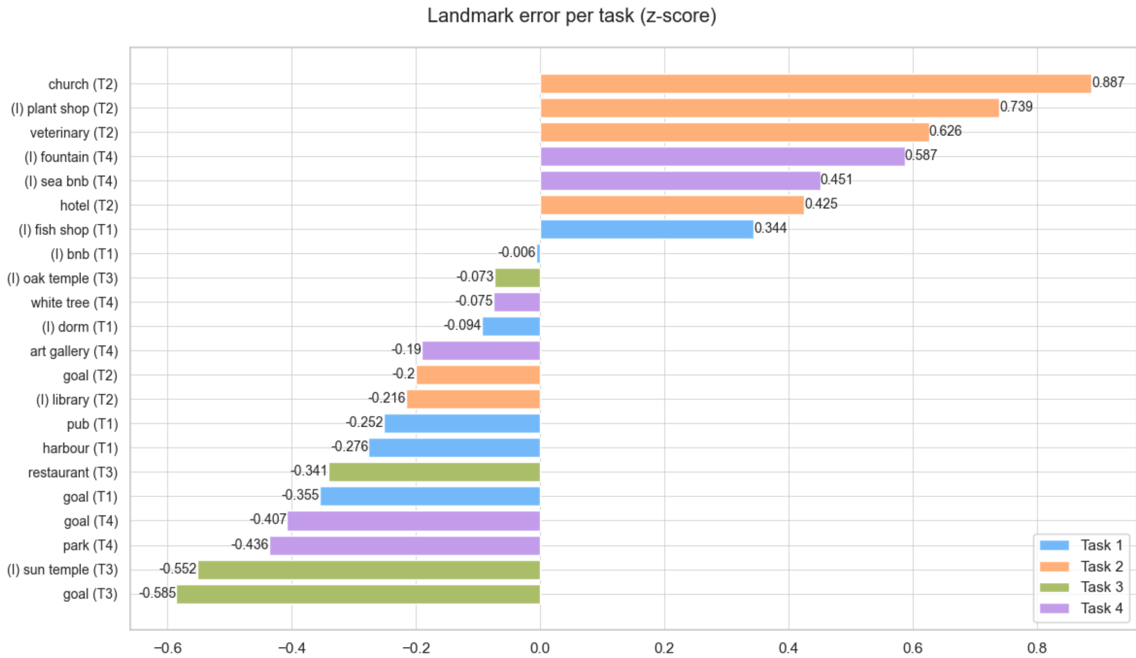


Figure 8: Horizontal bar plot, where each bar is the z-scored placement error for one landmark in a given task.

We can see the landmarks' names on the y-axis, along with their task number, also identifiable by the bar colour. Also, a `(I)` symbol next to the name indicates that the landmark sits on an intersection. We can see how the distribution varies. The two most difficult landmarks are in task 2, while the two easiest are in task 3, following the difficulty design. Most of the intersection landmarks are harder to recognise.

The plot in Figure 9, with a more immediate heatmap, shows how challenging the tasks are.

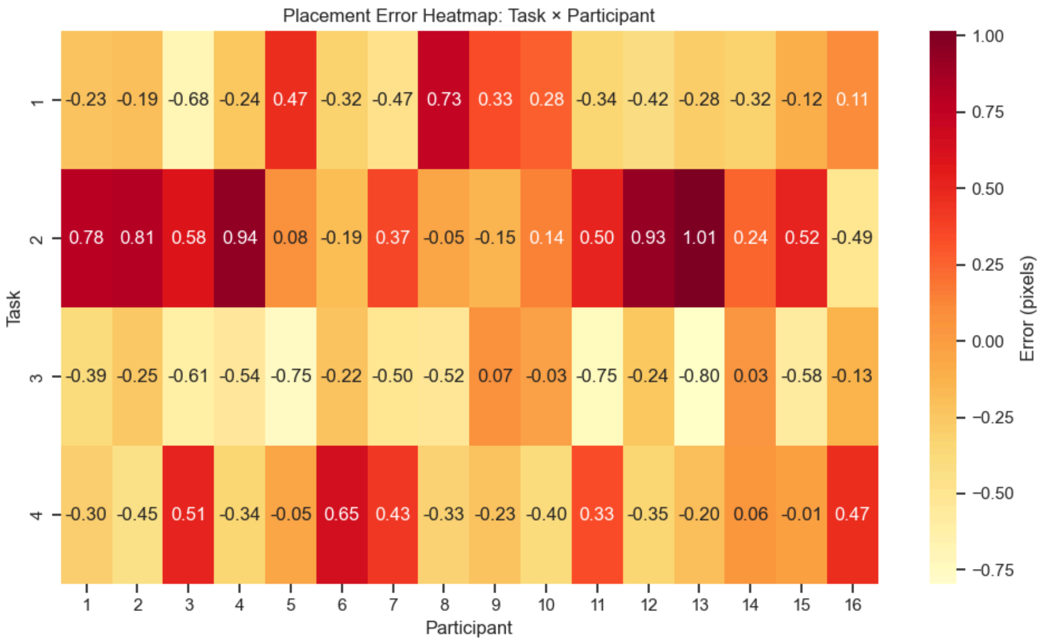


Figure 9: Average landmark placement error for a given participant (on the horizontal axis) and a given task (vertical axis). Each cell represents the z-scored average placement error for each participant on a given task. On the y-axis are shown the four tasks, while each value of the x-axis corresponds to a player. Lighter colours indicate lower error, while a more intense red indicates higher error when filling the sketch map with landmarks.

Finally, we can take a more general look with Figure 10, which observes the trends across tasks; it highlights each dependent variable's mean value for each Task and plots them all, stacked, in the last subplot. We can generally observe a higher spike on task 2, followed by a lower one on task 3, although it varies between variables, and there is no overall tendency.

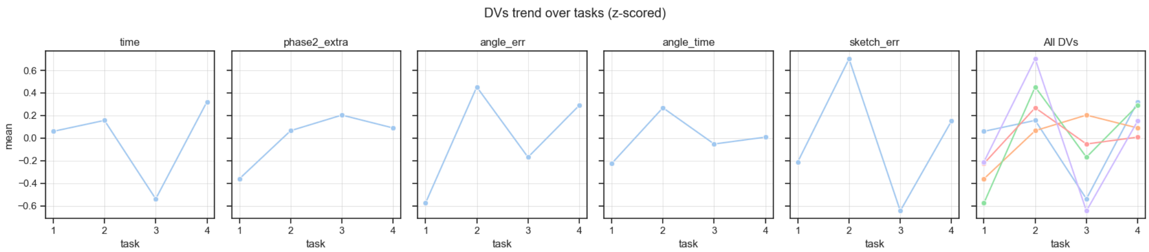


Figure 10: Dependent variable mean values across tasks.

4.2. Regarding performance

In this section, we go through each Dependent Variable's results. In each plot, tasks are laid out on the x-axis, and two different lines, one for each interface, describe the data; colours are consistent throughout the whole report, showing LF in orange and TBT in blue. Raw data points are plotted as small dots around the task index.

In Table 4, we report the results of the iterative model selection process we discussed in Section 3.5. All the models consider a random intercept, although it is not reported in the summary.

<table>
<tr><th>Dependent variable</th><th>Summary</th><th>Formula</th></tr>
<tr><td>Learning phase time</td><td>M4: Interface x Task, Rand. Slope on Interface</td><td><code>phase1_time ~ Iface * Task + (1 + | Iface | participant)</code></td></tr>
<tr><td>Recall phase time</td><td>M4: Interface x Task, Rand. Slope on Interface</td><td><code>phase2_extra ~ Iface * Task + (1 + | Iface | participant)</code></td></tr>
<tr><td>Landmark-placement error</td><td>M3: Interface x Task)</td><td><code>sketch_err ~ Iface * Task + (1 | participant)</code></td></tr>
<tr><td>Angle time</td><td>M1: Interface effect only</td><td><code>angle_time ~ Iface + (1 | participant)</code></td></tr>
<tr><td>Angle error</td><td>M3: Interface x Task</td><td><code>angle_err ~ Iface * Task + (1 | participant)</code></td></tr>
</table>

Table 4: Summary of model selection process results, showing the selected model for each dependent variable. The formula syntax follows the one in *lme4* package for R.

For reference, a plot of the diagnostics for the first model in Figure 11 is provided, including Residuals vs Fitted, a Q-Q plot of residuals, a scale-location plot, and a caterpillar plot. This set of plots aids understanding of how our model fitted the data.

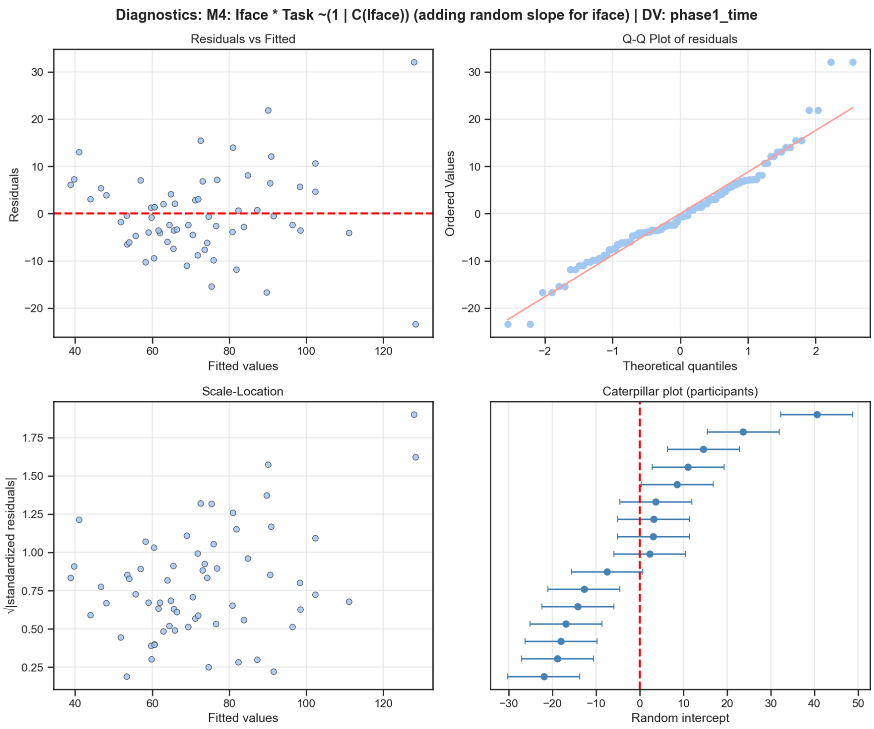


Figure 11: Collection of diagnostic charts regarding fitting process for learning phase travel time. From left-top to right-bottom: residuals vs fitted values, quantile-quantile plot of residuals, scale-location plot of residuals, and caterpillar plot for group variance.

The top-left chart is a scatter plot of the model residuals, on the vertical axis, against the predicted values, on the horizontal axis. For correct use of LMMs, we expect linearity, i.e., having residuals randomly scattered around zero and not forming any specific pattern, and a constant variance, i.e., roughly the same spread of residuals across the line. In this case, linearity holds, except for two outlying points after fitting = 120; the spread is similar across the fitted range, with more extreme residuals at high fitted values.

The top-right chart is a quantile-quantile plot of the model residuals against a theoretical normal distribution. If residuals were normally distributed, as assumed by LMMs, the points would lie on a straight line. Looking at the plot, we can see that most points lie relatively close to the straight line, with tails in the top-right and top-left that match the previously observed outliers.

The scale-location plot, in the bottom-left, helps check for heteroscedasticity, i.e., uneven variance in the residuals. It plots the square root of the absolute standardised (z-score) residuals on the vertical axis against the fitted values on the horizontal axis. If the residuals' variability does not depend on the fitted values, we expect a horizontal cloud of points, which is what we observe in our plot.

Finally, the caterpillar plot on the bottom-right shows the estimated random intercept for each participant. The dashed vertical line marks the population mean intercept. It helps visualise how many participants differ from each other, and makes it quick to spot participants with extreme random intercepts. In our case, intercepts vary widely, from -30 to $+45$, indicating that participant-level variability is an important source of variance, which justifies our use of random intercepts in our models.

The entire set of diagnostic plots for each variable is available in Section A.3.

Note on simple effects Although the Wald tests for the interaction effect for *learning time* and *landmark placement error* did not reach statistical significance, we still report their simple effects analyses. The global tests of interaction are typically conservative, especially in models with multiple parameters and limited statistical power. Our small sample size increases the likelihood of Type II error, meaning that the absence of a significant overall interaction does not necessarily imply the absence of meaningful differences at specific levels. Thus, we provide these results to give a more detailed perspective on the observed patterns, while acknowledging that such findings should be interpreted cautiously and considered exploratory in the absence of a significant global interaction.

Table 5 summarises the results of Wald's tests ran for each variable, indicating which effect is significant.

	Interface	**Task**	Interface x Task
Learning Time	$\boldsymbol{\chi^2 = 18.76}$ $\boldsymbol{p = 0.0003}$	$\boldsymbol{\chi^2 = 60.40}$ $\boldsymbol{p < 0.0001}$	$\chi^2 = 6.99$ $p > 0.05$
Recall Time	$\boldsymbol{\chi^2 = 13.11}$ $\boldsymbol{p = 0.004}$	$\boldsymbol{\chi^2 = 21.86}$ $\boldsymbol{p < 0.0001}$	$\boldsymbol{\chi^2 = 13.62}$ $\boldsymbol{p = 0.003}$
Landmark pl. error	$\chi^2 = 1.92$ $p > 0.05$	$\boldsymbol{\chi^2 = 18.25}$ $\boldsymbol{p = 0.0003}$	$\chi^2 = 6.05$ $p > 0.05$
Pointing error	$\chi^2 = 0.06$ $p > 0.05$	$\boldsymbol{\chi^2 = 13.43}$ $\boldsymbol{p = 0.003}$	$\chi^2 = 1.69$ $p > 0.05$
Pointing time	$\chi^2 = 0.061$ $p > 0.05$	-	-

Table 5: Results of Wald's tests showing main effects of interface, task, and interaction between the two, for each variable. χ^2 and p-values are reported. Note how the task effect is significant in every measure, while the interface effect is only for travel times.

4.2.1. Learning phase time

For the learning-phase travel time, the selected model was M4, with task-interface interactions and a random slope for the interface.

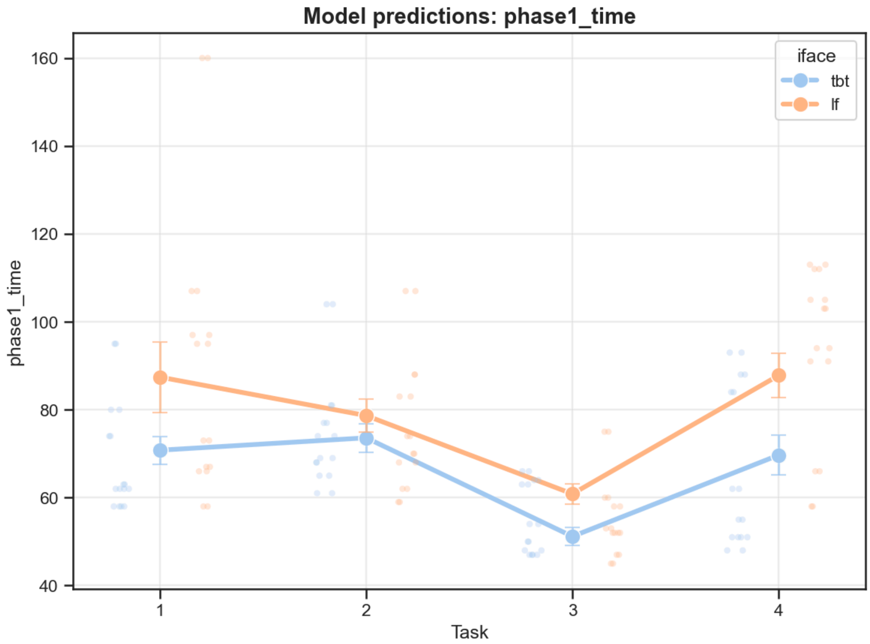


Figure 12: Navigation time during the learning phase, predicted by the model and with standard error lines.

Task	TBT Effect	SE	z	p
1	−16.62s	4.40	−3.77	**0.0001**
2	−5.05s	4.40	−1.14	0.25
3	−9.68s	4.42	−2.19	**0.028**
4	−18.13s	4.41	−4.11	**0.00004**

Table 6: Simple effects of interface on learning time for each task. Negative values indicate TBT requires less time. Wald's test shows that the overall interaction effect is not significant.

As shown in Figure 12, TBT is significantly faster than LF ($\chi^2 = 18.76$, $p = 0.0003$). Task difficulty matters ($\chi^2 = 60.40$, $p < 0.0001$), while the interaction between task and interface is not significant ($\chi^2 = 6.99$, $p = 0.072$). Simple effects in Table 6, however, hint to a significant impact of TBT interface on tasks 1 (Coeff.$= -16.62$, SE$= 4.40$, $z = -3.77$, $p = 0.0001$), 3 (Coeff.$= -9.68$, SE$= 4.42$, $z = -2.19$, $p = 0.028$), and 4 (Coeff.$= -18.13$, SE$= 4.41$,W$z = -4.11$, $p = 0.00004$). One single outlying value is found in interface LF, task 1. A detailed description of how outliers are defined can be found in Section 3.5.1.

4.2.2. Recall phase time

For the recall phase travel time, the selected model was also M4, with task-interface interactions and a random slope on the interface.

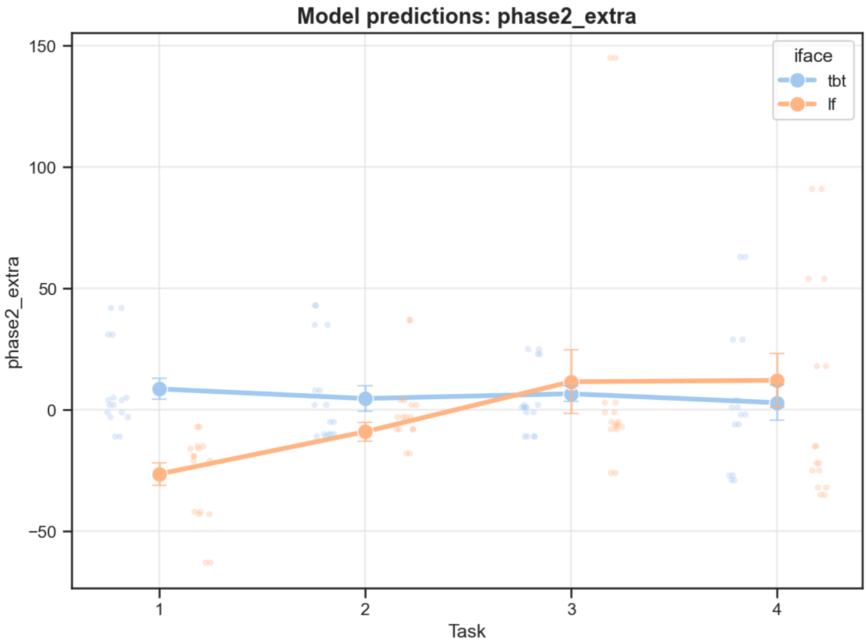


Figure 13: Additional time travelled in unaided return phase, predicted by the model, and with standard error lines.

Task	TBT Effect	SE	z	p
1	+35.08s	11.30	3.11	**0.002**
2	+13.71s	11.39	1.20	0.228
3	−4.97s	11.34	−0.44	0.662
4	−9.20s	11.38	−0.81	0.419

Table 7: Simple effects of interface on recall extra time for each task. Positive values indicate TBT requires more time. Wald's test shows that the overall interaction is significant.

Interaction	Coef.	SE	z	p
Intercept	−26.455	8.831	−2.996	**0.003**
C(iface)[T.tbt]	35.081	11.297	3.105	**0.002**
C(task)[T.2]	17.416	9.507	1.832	0.067
C(task)[T.3]	38.048	9.414	4.042	**0.000**
C(task)[T.4]	38.605	9.751	3.959	**0.000**
C(iface)[T.tbt]:C(task)[T.2]	−21.368	13.166	−1.623	0.105
C(iface)[T.tbt]:C(task)[T.3]	−40.047	13.112	−3.054	**0.002**
C(iface)[T.tbt]:C(task)[T.4]	−44.285	13.505	−3.279	**0.001**

Table 8: Interaction effects on recall extra time relative to the reference Task 1 LF.

The plot in Figure 13 shows a crossover interaction between interface and task, and is supported by Wald's tests ($\chi^2 = 13.62$, $p = 0.003$), indicating that the interface effect varied across task difficulty. Inspecting simple effects in Table 7, we can see a trend starting with an advantage for LF in task 1 (Coef.$=35$, $z = 3.11$, SE$=11.29$, $p = 0.002$). This trend decreases in task 2 (Coef.$=$

13.71, $z = 1.20$, SE$= 11.38$, $p = 0.228$) and reverses in tasks 3 (Coef.$= -4.9$, $z = -0.43$, SE$= 11.34$, $p = 0.661$) and task 4 (Coef.$= 9.2$, $z = -0.80$, SE$= 11.38$, $p = 0.418$), although these individual differences did not reach significance. Looking at Table 8, the interaction terms for task 3 and task 4 were significant (Coef.$= -40.047$, $z = -3.054$, SE$= 13.112$, $p = 0.002$; Coef.$= -44.285$,W$z = -3.279$, SE$= 13.505$, $p = 0.001$), confirming the crossover relative to task 1. We identified one outlier in the LF interface for task 3.

4.2.3. Sketch error

For landmark-placement error, measured in pixel distance, the selected model was M3, with task-interface interactions but no random slopes.

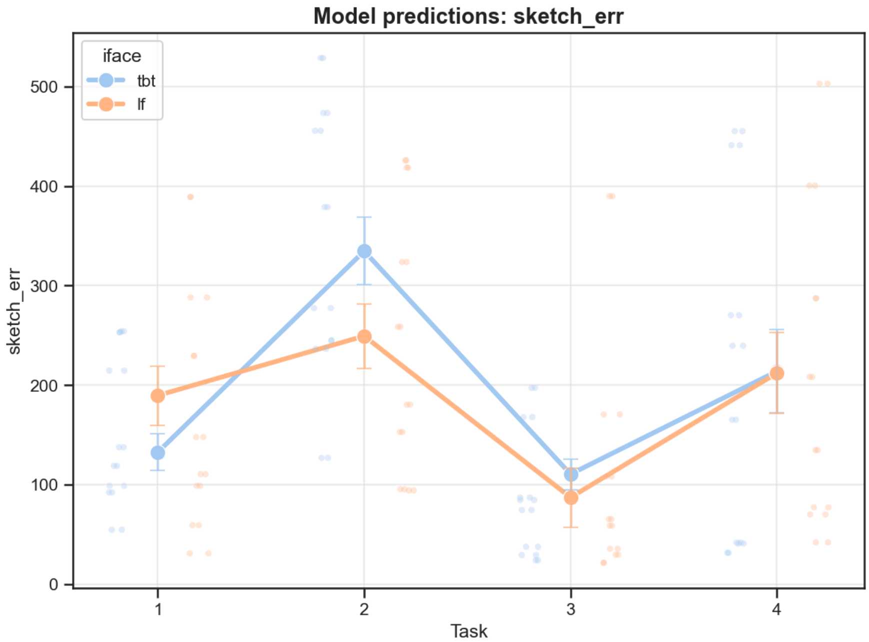


Figure 14: Landmark placement error, predicted by the model, and with standard error lines.

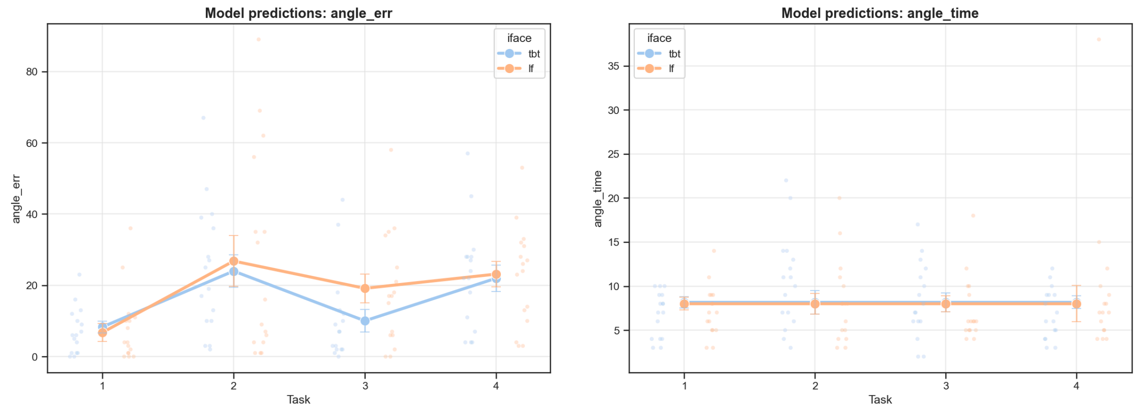
Task	TBT Effect	SE	z	p
1	−56.96px	41.07	−1.38	0.16
2	85.33px	40.83	2.08	**0.03**
3	23.40px	41.15	0.56	0.56
4	1.76px	40.80	0.04	0.96

Table 9: Simple effects of interface on landmark placement error for each task. Negative values indicate TBT was more accurate during the placement procedure. Wald's test shows that the overall interaction is not significant.

In Figure 14, the dominant effect is driven by task difficulty ($\chi^2 = 18.26$, $p = 0.0004$). No interface effect ($\chi^2 = 1.92$, $p = 0.166$) or interaction ($\chi^2 = 6.05$, $p = 0.109$) reach significance. We can see a high peak in task 2 and a low one in task 3, similar to the trend observed in Figure 12 for learning-phase travel time. Tasks 3 and 4 show tiny differences. Examining simple effects in Table 9, we recognise a significant TBT peak in task 2 (Coeff.$= 85.33$, SE$= 40.83$, $z = 2.08$, $p = 0.03$).

4.2.4. Angle measures

We now consider the variables for pointing error and decision time. For the former, the selected model was M1, which considers only the interface; for the latter, M3 was selected, thereby studying interactions between interfaces and tasks.



Pointing angle error, measured in degrees | Pointing decision time, measured in seconds

Figure 15: Angle measures predicted by the model and with standard error lines.

As shown in Figure 15, decision time measurements are not really informative and do not differ across interfaces, as supported by Wald's test ($X^2 = 0.061$, $p = 0.8$).

Regarding the pointing error, we do not observe significantly different values across the interfaces ($X^2 = 0.066$, $p = 0.797$). The error is primarily driven by task difficulty ($X^2 = 13.439$, $p = 0.003$); in particular, tasks 2 and 4 show the largest errors, while task 3 isWthe easiest, especially for TBT participants. No significant interaction between task and interface is found ($X^2 = 1.69$, $p = 0.63$), and no simple effects are reported either, as they were uninformative.

4.2.5. Correlating performance variables with self-reported differences

Let us examine the impact of individual differences on the dependent variables. We can see Spearman's correlation coefficient plotted onto a heatmap in Figure 18.

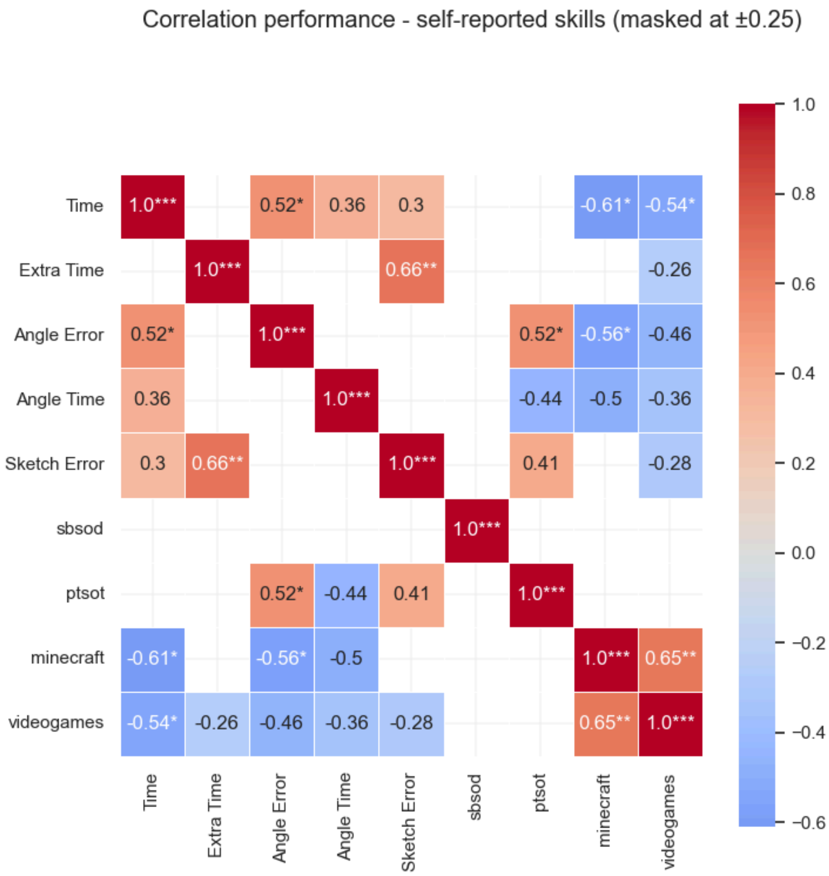


Figure 18: Masked heatmap reporting correlation coefficients between dependent variables and self-reported skills. Significance level is reported with one star ($p < 0.05$), two stars ($p < 0.01$), or three stars ($p < 0.001$).

Weak correlations between $r = -0.25$ and $r = 0.25$ were masked for simplicity of reading. The whole unmasked plot and the significance levels are available in Section A.4.

As concerns dependent variables, we observe moderate positive correlations between learning phase time and angle error ($r = 0.52$, $p = 0.04$), angle time ($r = 0.36$, $p > 0.5$), and sketch error ($r = 0.3$, $p > 0.5$), though the latter two do not reach significance. Moreover, as recall phase time grows, sketch error grows as well ($r = 0.66$, $p = 0.005$).

Sense of direction does not show any strong correlation in general. PTSOT shows moderately low correlations with sketch error ($r = 0.41$, $p > 0.5$), angle time ($r = -0.44$, $p > 0.5$), and angle error ($r = 0.52$, $p = 0.039$), the only significant one.

Stronger interactions are found between experience in virtual environments and our measurements. Trivially, Minecraft and video game experiences are strongly correlated ($r = 0.65$, $p = 0.006$). Both show negative correlations with learning time ($r = -0.61$, $p = 0.012$ and $r = -0.54$, $p = 0.029$), angle error ($r = -0.56$, $p = 0.023$ and $r = -0.46$, $p > 0.5$), and angle time ($r = -0.5$ and $r = -0.36$, although both $p > 0.5$). Video game experience is weakly negatively correlated with recall travel time ($r = -0.26$) and sketch error ($r = -0.28$), but neither shows significance.

Finally, we can see histograms with the distribution for self-reported skills in Figure 19.

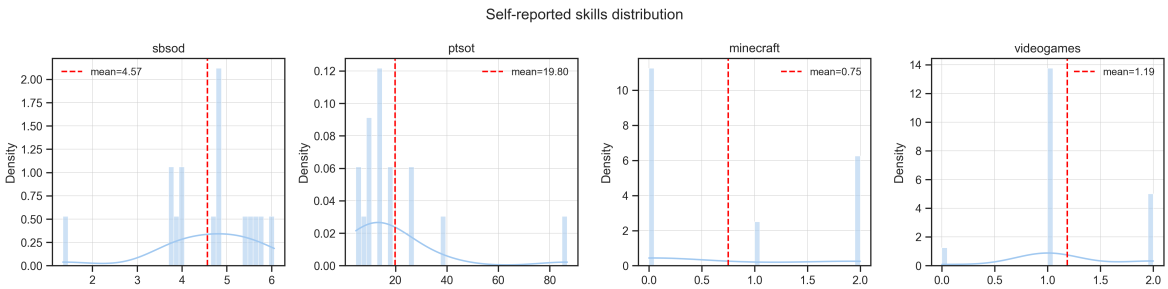


Figure 19: Density distribution and mean value for each self-reported skill.

	mean	std
SBSOD	4.568	1.141
PTSOT	19.805	19.923
Minecraft	0.75	0.93
Videogames	1.187	0.543

The SBSOD distribution, representing sense of direction, is skewed towards higher scores, with a mean of 4.57. The second plot, for PTSOT, showing average angle error, is skewed towards a lower error on the left, with a mean of 19.80. While experience with Minecraft is quite split between “never played” and “play often”, experience with video games seems to follow a normal distribution centred at value 1.19, which stands for “some experience with video games”.

4.3. Regarding Gaze

We conducted an exploratory investigation of fixation rates and durations, although without advanced statistical techniques. Each subplot shows data from one task, with a multiple-bar plot for LF data on the left and TBT data on the right.
Each bar describes an Area of Interest, namely *env* for the environment, *minimap* for the interface, and *direction* for the navigation cues shown above the minimap.

Please note that direction cues are not supposed to appear in LF, as they are not part of the interface, but a bug in the code sometimes triggers a direction box to appear there. It has happened very few times, as proved by Figure 20, which shows a tiny percentage; the larger bars in Figure 23 refer to fixation duration, not number, and indicate that participants glanced for some time at the appearing cues. The effect of this bug is likely negligible.

4.3.1. Fixation ratio

As an illustrative plot, in Figure 20, we report the ratio of fixation number for each AOI, divided by task.

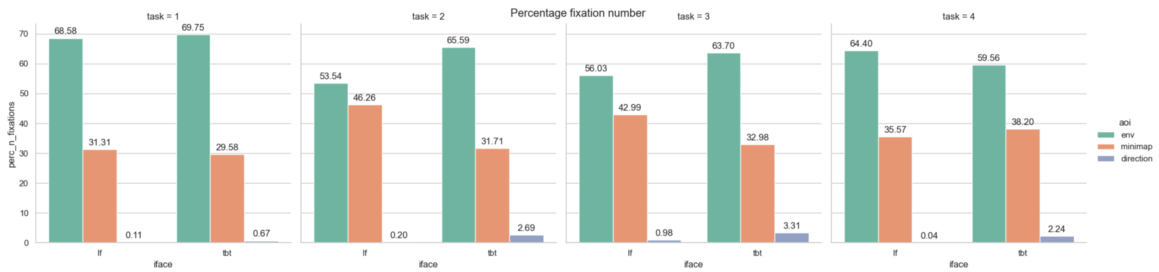


Figure 20: Ratio of fixations on a given AOI, divided by task (subplots) and interface (horizontal axis). Values are expressed in percentage related to a given interface.

For task 1, the ratios are almost identical, signalling no difference between interfaces. In tasks 2 and 3, we observe that using the TBT interface led to greater time spent looking at the environment than LF. This trend reverses in task 4, where LF users pay slightlyWmore attention to the environment than to the navigation interface.

4.3.2. Number of fixations

In Figure 21, we observe the number of fixations in each AOI, divided by interface and task. Fixations on the environment prevail, as previously shown. Table 11 reports the Wald's test values run for the selected model, consisting of the main effects of interface, task, and AOI, and their interactions. For all the following analyses, navigational cues were removed from the count.

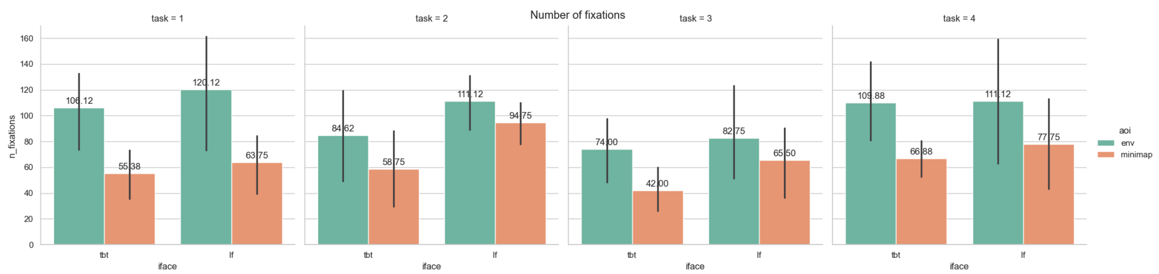


Figure 21: Number of fixations reported by AOI, task, and interface. Black lines indicate the standard deviation.

	χ^2 and pvalue
Intercept	$\chi^2 = 423.40$, $\boldsymbol{p < 0.00001}$
iface	$\chi^2 = 1.9034$, $p = 0.16$
task	$\chi^2 = 44.1449$, $\boldsymbol{p < 0.0001}$
aoi	$\chi^2 = 164.9804$, $\boldsymbol{p < 0.0001}$
iface:task	$\chi^2 = 9.5534$, $\boldsymbol{p = 0.02}$
iface:aoi	$\chi^2 = 16.9147$, $\boldsymbol{p = 0.00003}$
task:aoi	$\chi^2 = 50.7125$, $\boldsymbol{p < 0.0001}$

Table 11: Main effects and interactions between interface, task, and area of interest, tested with Wald's chi-squared test.

The results in Table 11 show a significant effect of task ($\chi^2 = 44.14$, $p < 0.0001$) and area of interest ($\chi^2 = 164.98$, $p < 0.0001$) on the number of fixations, contrary to the interface ($\chi^2 = 1.90$, $p = 0.16$). All interactions are significant. Interface modulates the effect of task ($\chi^2 = 9.55$, $p = 0.02$) and AOI ($\chi^2 = 16.91$, $p = 0.00003$), and the latter two also interact with each other ($\chi^2 = 50.71$, $p < 0.0001$).

AOI	LF	TBT	LF/TBR ratio	SE	z ratio	p-value
Environment	84.8	74.0	1.15	0.028	5.36	$\boldsymbol{p < 0.0001}$
Minimap	59.9	44.3	1.35	0.042	9.58	$\boldsymbol{p < 0.0001}$

Table 12: Interaction of interfaces on AOIs, expressed in LF/TBT. A higher ratio indicates a higher number of fixations from LF.

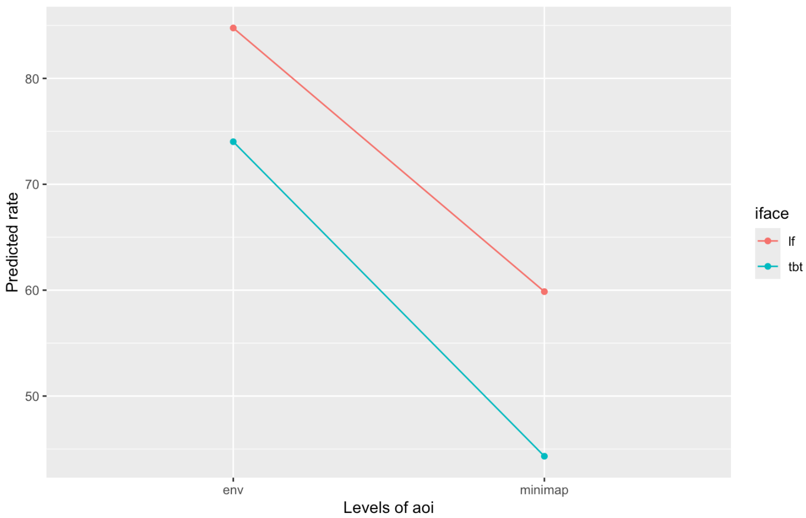


Figure 22: Impact of interface on Area of Interest studying number of fixations.

Table 12 shows the effects of the interface on AOI, calculated with the `emmeans` R package, and also plotted on Figure 22. LF produces a larger number of fixations than TBT both on the environment (15% more, $z = 5.36$, $p < 0.0001$) and on the minimap (35% more, $z = 9.58$, $p < 0.0001$). The complete output from R is available in Section A.5.

4.3.3. Fixation duration

In Figure 23, we see the average fixation duration for each AOI, divided by Task.

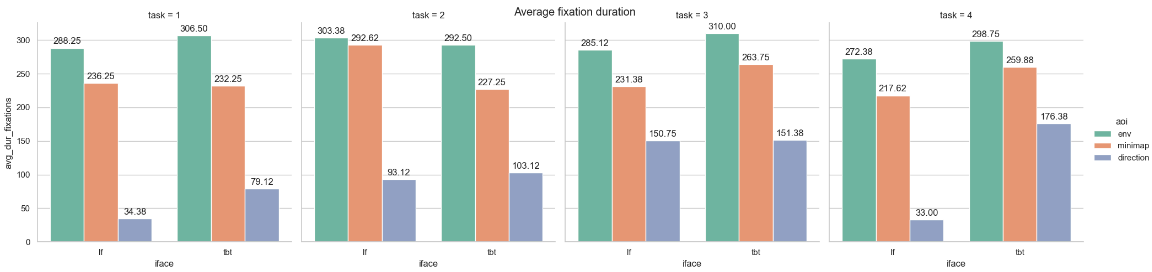


Figure 23: Average fixation duration in ms, divided by task (subplots), interface (horizontal axis), and AOI.

On average, fixations are longer on the environment compared to the interface. The ratios change across tasks. Apart from task 2, in which the values are closest to the rest, participants fixate longer on the environment when using the TBT interface. Their fixations last longer in general than when using LF. Fixation length on the minimap varies quite a bit across tasks: in task 1, it is reasonably similar across interfaces; in task 2, it is longer in LF; but in tasks 3 and 4, the trend reverses, showing longer fixations on the minimap for TBT.

When running Wald's test on model M1 to test the effect of interface, task, and AOI on fixation duration, only AOI reached a significance level. This is expected, as AOIs are purposely different parts of the screen that require different engagement by the user. No further analysis was done.

5. Discussion

This chapter discusses the relevant points from the methods and results. First, we examine the results shown in Section 4 and link them to our original hypotheses. Next, we go through crucial design decisions introduced in Section 3. We reflect on our strengths and weaknesses and discuss the possibilities this study opens for future work.

5.1. Examining results

Let us outline what the expected results are derived from the hypotheses stated in Section 1.3.

For the learning phase in each task, we expect the *turn-by-turn* interface to show shorter completion times than the *landmark-focused* interface. Participants can rely on the blue pathline to reach the destination, potentially without actively encoding the environment as LF users do. If that were consistently not the case, the TBT interface would have a fundamental design issue, making it slower than LF. Drawing on the navigation performance component of Hypothesis 1, we expect the recall phase to be shorter when the Task is played with LF than with TBT. Finally, considering survey knowledge variables, we expect landmark placement error on the sketch map to be lower after playing LF than after playing TBT. A similar behaviour is expected for the pointing angle error and pointing time.

In this section, we review and discuss the results shown in Section 4, starting with Task difficulty, then moving on to performance metrics, and finally closing with an exploratory gaze analysis.

5.1.1. Task difficulty

Let us now discuss whether the results regarding task difficulty are consistent with our original design.

Additional task-difficulty scales were calculated post hoc based on players' performance to verify alignment with the original design values. We used performance data (raw and without outliers) and landmark placements. The goal was to understand if our original design was valid. Regarding performance data, we aggregated each DV per task, calculated the mean, and ranked tasks for each variable. We then averaged across columns (axis=1) and divided by the max value to obtain a difficulty score between 0 and 1. The exact process was the same for sketch, but this time aggregating landmark placement accuracy.

task	original design	raw data	without outliers	landmark error	**average**
1	0.75	0.625	0.625	0.50	0.625
2	1.00	0.875	0.938	1.00	0.953
3	0.60	0.562	0.667	0.25	0.520
4	0.75	1.000	0.750	0.75	0.812

Table 13: Original design task difficulty ranking along with post-hoc calculated scales, based on every variable (raw data and without outliers) and only on landmark placement error.

As shown in the resulting Table 13, all scales' rankings follow the original order: task 3 (the easiest), then task 1, task 4, and finally task 2 (the hardest). While our original rank placed tasks 1 and 4 at the same level, other scales agree that task 4 is more challenging. The last column shows all the columns averaged into one single ranking. Overall, this result seems to confirm the scale we initially designed and empirically shows differences in task difficulty.

More detailed representations of this data are visible in Figure 8, Figure 9, and Figure 10.

The first plot highlights the specific difficulty of landmarks, as measured by average placement error. Unsurprisingly, locations from task 2 dominate the chart, while two from task 3 make up the very bottom. The rest of the distribution is harder to associate with a single trend or task, as it is pretty variable. It could indicate that landmarks are not inherently more difficult than other landmarks, based on their own features. We could also be missing a pattern and not recognise a trendWin a not explicitly stated landmark feature (e.g., if they are linearly organised by distance from the water, given its relevance as a global landmark).

The heatmap in Figure 9 underlines the differences across tasks. Specifically, we can observe that task 2 has a more intense red colour, while task 3 is the lightest of the group, reinforcing the exceptionality of these two tasks. We can also see that no participant had an all-around outlying performance; i.e., no one performed very poorly or very well across all the tasks; this indicates that tasks are balanced in difficulty, but also underscores once again that individual differences matter a lot: people, on average, perform worse on the 2nd task and better on the 3rd, but performance varies quite much for the others.

Finally, we have a general overview of the trends in Figure 10. We see one performance metric per subplot, while the last one stacks them all. We cannot see a clear emerging pattern across all DVs, although two spikes are noticeable in tasks 2 and 3, as previously mentioned.

5.1.2. Performance results

This section is dedicated to the results from LMMs, our primary analysis. Performance variables differ in their meaning and outcomes, yielding contrasting results that do not allow us to draw a clear conclusion. Part of the reason is the limited number of participants. As a confirmation, when fitting several models with `statsmodels`, we received warnings about the random-effects covariance being singular. This message indicates that the model tried to estimate variance/covariance parameters that the data do not support. A common cause is a random structure that is too complex for the number of groups (16 in our case), leading to unreliable estimation of random slopes and intercepts. Still, all of our models converged, and we inspected diagnostic plots, reported in Section A.3, as a sanity check procedure.

We now proceed to discuss the plots described in Section 4.2.

In both learning and recall travel time variables, the chosen model was M4 with a random slope for the interface, suggesting that the interface effect varies across participants. Some players might be more advantaged by LF for various reasons, including differences related to their self-reported sense of direction and spatial orientation, as we discuss in Section 5.1.3; experienced video gamers, for instance, could have an advantage in wayfinding in virtual environments (Yavuz et al., 2024).

Learning phase, travel time Our guiding question for the *learning phase time* variable is whether, during phase 1, TBT users arrive at the end goal faster than LF users. Our TBT interface was designed to benefit users by providing a clear path to follow, further aided by their familiarity with standard commercial interfaces. Therefore, we expected it would take less time to reach the destination than LF. This hypothesis is consistent with what we see in Section 4.2.1: TBT is significantly faster than LF in the first phase, and the task difficulty also has an impact on the performance, with the largest results in tasks 1 and 4. Overall, this suggests that the interface we designed effectively increases navigation efficiency during the learning phase, as commercial navigation apps do.

Recall phase, travel time As concerns the *recall phase time* variable, we are interested in whether, during phase 2, LF users return to their starting point faster than TBT users. The LF interface was designed to encourage users to pay more attention to the environment during the learning phase, as they receive no other aid, unlike TBT participants. We expected participants using LF to gain better route knowledge of the environment and to be advantaged in the recall phase. Examining the data in Section 4.2.2, we see promising results. According to Wald's tests, both interface and task difficulty effects are significant, meaning that they do have an impact on recall time. The most significant result, as shown in Table 7, is an additional time of 35s for TBT users in task 1 compared to LF. The effect depends on the interactions (Wald: $\chi^2 = 13.62$, $p = 0.003$), as we can see from the crossover pattern in Section 4.2.2 and Table 8; LF is faster, as predicted, during tasks 1 and 2, but then TBT overtakes it in tasks 3 and 4, giving us contrasting results. The impact of task difficulty can change the direction of the interface effect.

Landmark placement error and angle measurements Our guiding question when analysing *landmarks placement error* in Section 4.2.3 is whether LF users are more precise in recalling landmark positions than TBT users. In the case of Section 4.2.4, instead, we asked whether LF users are more accurate and faster at pointing to their origin point than TBT users. We expected our landmark-focused interface to encourage survey knowledge acquisition by requiring users to navigate using landmarks. Let us start by identifying pointing time as uninformative, suggesting discarding the variable in future work. Its chosen model is M1, only considering interface effect. No significant or informative value is found in the analysis, as visible in Section 4.2.4. For sketch and angle error, the chosen model was in both cases M3, indicating that interface random slopes added no value and that the interface does not depend on user differences.
Performance is better explained by the tasks in these two instances, as task difficulty is the only main significant effect found. Looking at the plots, we observe similar trends across interfaces, with landmark accuracy standing out for a high peak in task 2 and a low peak in task 3, similar to the learning-phase time seen in Section 4.2.1. For the former measure, we should take into account the unit of measure, pixels. Minor errors could even be due to inaccurate distance measurements during data processing. Task 3 seems to be reaching ceiling-level, likely since it only has four landmarks. The angle measure is more consistent and shows smaller differences, although it also presents a low peak in task 3. These results reject our first hypothesis, as there is no clear advantage of the LF interface over the TBT.

General overview

We find two main meaningful results from our linear mixed models.

The first one is the task difficulty impact, which dominates across variables, indicating significant differences between tasks. Task 2 shows a higher angle and sketch error compared to other tasks, and it does not reach significance when studying main effects in learning and recall time. Task 3 seems to be consistently the easiest one in the group: the fastest in learning time, ceiling level in recall time and angle error, and the most accurate in landmark placement. Tasks 1 and 4 tend to be in between the former two, not following a precise general trend. A particular case is task 3, which showed minimal differences across measures (recall time, sketch error, pointing error), although this ceiling effect was not anticipated during task design: our difficulty scale predicted it to be moderately easy; this is discussed in Section 5.3.4, concerning flawed task difficulty design.
This is a successful outcome for our task design process, as we effectively introduced challenging elements at different levels. It would be an interesting future perspective to investigate further which factors have the most significant impact on difficulty and how they interact.

The interface effect shows more complex patterns. Our initial hypothesis predicted that LF would yield more accurate survey spatial knowledge (H1: landmark placement accuracy, pointing angle error, pointing decision time), but it falls short when considering the collected data. In-

terface does not significantly impact the aforementioned measures. However, it does on our route knowledge variables: learning and recall travel times. TBT has a beneficial effect on phase one, making participants travel faster thanks to the helpful aid. Interface also has a significant effect on the recall phase, although the direction does not completely align with our expectations: LF is helpful in tasks 1 and 2, but not in 3 and 4. As the task interacts with the interface, it is plausible that certain features of tasks 1 and 2′s structures advantage LF, but we do not have enough data to exactly pinpoint a particular characteristic.
Our results seem to follow the stages of development of spatial knowledge indicated in the dominant framework, which sees route knowledge form earlier compared to survey (Siegel & White, 1975). This could explain why we see an interface effect on route knowledge measures (travel times) and not on survey knowledge ones, which might require a longer experiment or repeated trials to develop. Finally, the crossover pattern in the recall phase travel time is to be further investigated in future work.

5.1.3. Correlation between DVs and self-reported skills results

Let us consider the correlation matrix shown in Figure 18. Some of the measures are not significant, but we still report them as they can be an interesting starting point for future work.

Performance variables A longer time to complete a task is associated with larger pointing error ($r = 0.52$), suggesting a correlation between navigation performance and survey knowledge, and may reflect a reciprocal relationship. Angle error and angle time are not strongly correlated. We had two competing hypotheses about this association: on one side, we expected a higher decision time to be related to precision, as users accurately move their cursor to catch the correct direction perfectly; on the other hand, an opposite explanation can also be given, arguing that a player with a good mental map and clear idea of their arriving direction can confidently quickly select the correct angle. In this sense, given the inconclusive data, this small correlation suggests that both strategies may be present across participants. In future work, a more thorough investigation of individual differences should be undertaken to understand which hypothesis prevails. A larger sample of data would also be beneficial to make more clear possible effects that remain now hidden clearer.

SBSOD and PTSOT We did not find any significant correlations with SBSOD scores. In literature, SBSOD is generally more strongly correlated with measures of spatial knowledge acquired from direct experience in the environment than with those acquired from maps, video, or virtual environments, which may explain our findings. Concerning PTSOT, we find that higher values (i.e., worse mental rotation) are associated with higher angle error ($r = 0.52$); correlation with sketch error ($r = 0.41$) and angle time ($r = -0.44$) is also considerable, but they did not reach significance. It is plausible that poor spatial orientation skills harm performance, consistent with results from the literature (Hejtmánek et al., 2018). It seems that the decision time for pointing depends on other factors, likely related to individual differences.

Minecraft and Video games experience Finally, Minecraft and videogame experiences generally have a positive impact on performance; this is consistent with previous work showing that videogame play is associated with enhanced spatial cognition (Yavuz et al., 2024). The more one plays Minecraft and/or video games, the less time it takes in general ($r = -0.61$, $r = -0.54$), and the better one's pointing skills (error: $r = -0.56$, $r = -0.46$; time: $r = -0.5$, $r = -0.36$), although only the association between Minecraft and angle error is significant. Video game experience has an impact on recall travel time ($r = -0.26$) and sketch error ($r = -0.28$), but neither reached significance. Finally, the two variables are positively correlated ($r = 0.65$), which is expected because Minecraft is itself a video game, thus a higher value in the former will carry a higher value on the latter.

General overview Looking at the histograms in Figure 19, we notice different data distributions for self-reported skills. SBSOD is skewed toward higher scores, indicating that our participants primarily reported a strong sense of direction. PTSOT data also looks generally positive, with an average angle error of 19.90° and only one considerable value on the tail. Finally, the Minecraft experience looks categorical, with many values split between "never played" and "play often"; the video games experience centres around "play sometimes", with few extreme values.

5.2. Gaze

Looking at the fixation ratios in Figure 20, we see that environment fixations dominate across all tasks and interfaces. The ratio depends on the task: in tasks 2 and 3, TBT users look more at the environment than LF; in task 4, the trend reverses; and in task 1, the numbers are almost identical. LF users seem to spend the same amount of time, or more, lookingWat their interface as TBT users do.
The chart seems to contradict our H2 prediction that TBT users would show more frequent fixations on the interface, while LF users would focus more on the environment.

To further investigate, we plotted the number of fixations in Figure 21 and analysed them with GLMMs. According to Table 11, the interface, per se, does not have a significant impact on the number of fixations, contrary to the task difficulty. However, there is an interaction effect between interface and area of interest, as we can see in Table 12 and Figure 22. The results show that LF users make more fixations both in the environment (15%) and in the interface (35%) compared to TBT.

This pattern suggests that LF navigation required a more active visual search in both the environment and the interface. Without the path line, participants probably were more drawn to frequently interact with the minimap to identify the next landmark on their path; mentally rotating the map orientation to match their egocentric perspective is also an effort that TBT users do not experience, as the path to follow is drawn. The 35% increase in minimap fixations for LF users positions in contrast to our original hypothesis (H2), in which we expected our landmark-focused interface to encourage a larger number of fixations on the environment compared to turn-by-turn.

Another observation we already mentioned in Section 5.3.6 concerns directional cues, whose fixations make up at most 3.31% of a task's total. Navigation cues require more effort than following a path line because they require converting textual instructions into mental visualisations of the world. An intuition for this is evident in Figure 23, which examines the average fixation duration. If participants were not using them at all, we would expect them to either never look in that AOI, or just shortly to glance, more reacting to an appearing stimulus than for actually reading; the average duration of fixation hints that participants are actually looking at them and trying to read the instructions, but probably only try it few times, as seen in Figure 20. The tradeoff between effort required and helpfulness is likely negative. An interesting future idea would be to investigate their use in commercial navigation apps.

5.3. Regarding methods

During the process of implementing the experimental design, testing it on real hardware, analysing the collected data, and interacting with participants, we realised that some decisions were more based on heuristics and assumptions than on documented behaviours.

Upon completion of the tasks, participants were encouraged to provide informal feedback on the procedure, task difficulty, general observations, and any struggles encountered. We verbally collected many opinions. A more formal after-study survey would have provided a more standardised and analysable set of answers.

5.3.1. Time

A general issue was the lack of time. The entire research spanned from August to December 2025, totalling five months. We spent the first two months researching the literature and the following two months designing the experiment, collecting, and analysing data; the last month was dedicated to writing this manuscript. A substantial amount of time was spent designing the interfaces, implementing them with Minecraft mods, and testing their functionality; building the tasks in Minecraft, both the pilot and the final design, also required time. Finally, recruiting participants and finding spots for them to come to the laboratory naturally adds up. This short time window pressured us to make assumptions and decisions without thoroughly investigating the literature to build a stronger foundation. It hindered us from recruiting more participants, collecting additional data, and running more advanced analyses.

5.3.2. Participants

The population sample in this study is quite limited, both in terms of number and of internal variation. In terms of anagraphic data, most participants were highly educated, mostly male (nine males, seven females), and aged 20-35 years. Despite being aware of possible gender effects on spatial navigation abilities (Munion et al., 2019), we did not control for these factors to avoid escalating complexity and to restrict participants' selection criteria.

The other significant limitation is the small sample size. The leading cause is the limited time available and the data collection process spanning about 2 months. Furthermore, participants would receive no monetary compensation for their time, which could discourage participation in an on-site experiment lasting 45 minutes to 1 hour. The impact of a low N cannot be underestimated in terms of statistical significance. No generalisation can be made, and evidence found can only be seen as hints to confirm further. Furthermore, in complex models such as LMMs, a small sample size can lead to overfitting and make it harder for the model to estimate variance consistently.

5.3.3. Trials assignment

The assignment of interface and tasks between players, as shown in Table 2, introduces some bias we were unable to control. We collected data from 16 participants, and each of them played four tasks: two with TBT and two with LF. This means that every task is played 16 times (once by each player) and both interfaces are played 32 times (twice by each player). Assignments are not completely counterbalanced for other factors, however. We can get a clearer idea of the problem by looking at Table 14, which is calculated from Table 2.

		1st pos.	**2nd** pos.	**3rd** pos.	**4th** pos.
TASK	1	4	5	4	3
	2	4	4	4	4
	3	4	3	4	5
	4	4	4	4	4
INTERFACE	TBT	11	6	8	7
	LF	5	10	8	9

Table 14: Number of times a given interface or task is played in a certain position. The position indicates the point in the sequence of task that a participant would play a given task.

Interfaces are not played the same amount of time across positions, ie, the sequence in which the four tasks are played. For example, TBT is played 11 times in the first position, while LF is played only 5 times. This means that users start with the turn-by-turn interface 68% of the time. A similar, but weaker, effect is seen for tasks. For example, task 1 is played 5 times in the second position, but only 3 times as the last one. The same applies to interface sequence: there are 21 instances where LF follows TBT, 19 where TBT follows LF, and eight where the same interface is played twice in a row.
Tasks were supposed to be distributed evenly across positions to avoid bias from "played-first" or "played-last", but the assignment was incorrectly calculated. We did not study potential correlations between task or interface sequence and performance, not only for time reasons but also because, with such a small N, it was unlikely to detect significant interactions. This issue must be addressed to ensure the robustness of future work.

Although a between-subjects design, i.e., splitting participants into two separate groups, one per interface, would have simplified the analysis, we purposely chose a within-subject approach because of our sample size constraints. With only 16 participants, a between-subjects design would have increased the risk that one condition would contain more or less skilled players in navigation and video game experience, introducing a significant bias. A within-subjects design provided a safer solution for individual differences, because each participant serves as their own control, increasing the sensitivity to detect researched effects. Our choice, however, can be more vulnerable to carryover effect, harm late tasks by design due to fatigue, and requires a more complex counterbalancing structure, which we failed to design correctly.

5.3.4. Design - task difficulty

The calculation of task difficulty relied on limited evidence from previous literature, due to the novelty of the experimental scenario and the restricted amount of time. It was based on *path length* (in blocks), *number of landmarks*, and *number of turns*. The procedure for calculating difficulty is detailed in Section 3.3.2. The reason to choose these factors is that a higher number of them involves more objects to maintain in memory (Bushinski & Redick, 2025; Miller, 1956), creating a more complex challenge for path recall in the second phase and for landmark recall after the task. However, no in-depth research was conducted before this decision, mainly due to time constraints. Additionally, no weights are specified for these factors, effectively assuming they all have the same impact on task difficulty.

The fundamental issue is that we do not ensure a consistent level of difficulty for our tasks. As we will examine further in the results in Section 5.1, in several cases, we would expect a similar trend across DVs following the task difficulty score, but we did not necessarily see that. Many other factors are likely to affect participants' performance and perceived difficulty. Some examples include route height difference and geometry, visual homogeneity, and road materials. A more homogeneous landscape can make it hard to distinguish the features in the path (Stankiewicz & Kalia, 2007; Wang et al., 2025), and task 3, set in the desert, presents a very similar *sandy* colour in the surroundings and in the building blocks; the landmarks, however, are designed to stand out thanks to their different colours: the Oak Temple trees provide some green, the Sun Temple has purple-coloured structures on the top, and the restaurant shows an Italian flag above it. Three of the tasks included a large body of water visible along the route, possibly serving as a global landmark to encode other landmarks relative to it (Liu et al., 2022). Finally, task 1 is the only one where the starting point is clearly visible from the ending point, which might give a clearer overall idea of the path back (Omer & Goldblatt, 2007).

5.3.5. Design - landmarks

Landmarks' choice in terms of location, level of detail, and building type could be informed by more accurate research, for example, by analysing whether certain structures or colours are more memorable. Landmarks we use to build cognitive maps are not necessarily semantically objective, but can be meaningful to different people for different reasons, for example, a related memory or some peculiarity.
Our task design followed a different approach. We started by choosing task locations in urban areas of the map outside large cities, smaller in size, and with peculiar landscapes. We then identified reasonably long routes and selected buildings that are distinctive and identifiable to most people. Some were already present on the map (eg. the BnB in task 1) and were kept as they were; the majority were present in some way already, but we adjusted their looking to make them more clearly noticeable and associable with a specific location (eg. we added boats on the harbour, a sun structure on top of the sun temple, or plants outside the plant shop); a very few were created from scratch (eg. the Park in task 4 replaced a stadium building). The decision on the features that make a place distinctive was within our discretion, which may have led us to favour landmarks that are more recognisable to us, based on our navigation skills, experience, and interactions with similar places.
A possible critique is that landmarks do indeed stand out, but their association with the minimap is conveyed through their names as text. The representation of landmarks on the interface has a substantial impact on navigation performance (Kapaj et al., 2024). For simplicity of implementation, we stuck with the default representation provided by Xaero's Minimap mod, but this can compromise landmarks recognition on the map sketching task: users could likely remember animals outside the Veterinary (task 2), as there are no others around the maps, but then struggle to associate them with a "veterinary".

We performed an additional post-hoc analysis of landmarks at intersections to understand whether they are correlated with recognisability, finding that, unexpectedly, many intersection landmarks, specifically seven out of nine, fell among the 50% worst accuracy. We can observe this in Figure 8, where landmarks at intersections are marked with a `(I)` symbol before their names. Intersections are decision points, moments during navigation when users can choose whether to keep or change their direction, and are crucial for mentally reconstructing the way back; landmarks placed at intersections tend to be recognised faster than others for their importance in navigation (Janzen, 2006). Before concluding that this is a substantial contradiction to existing literature, we must consider possible explanations. Our definition of intersection might not precisely align with that of a decision point in existing findings, particularly in virtual environments. A Minecraft reconstruction of a real city differs from a real city environment or any non-cube 3D game, with movement options that differ and thus potentially hinder the perception of an intersection as elsewhere. The result is interesting, but further research is required to assess the situation.

5.3.6. Design - interface

Our interfaces were designed with precision to achieve each a specific purpose. Our general aim was to create interfaces with a working design, assess their functionality in virtual environments, and then stimulate further studies on landmark-focused interfaces directly in the real world.

The landmark-focused interface is thought to be minimalist, showing only landmarks, as they carry enough information for a participant to reach their goal without any other aid. This design is confirmed to work by the fact that players complete the learning phase across tasks using it. Only providing landmarks encourages route knowledge, as users are able to then return to the starting point when the interface is hidden. Initially, the next landmark on the route was supposed to briefly animate (e.g., bounce) to aid the user in deciding which landmark to move towards; this option was discarded, though, as the animation was too distracting from the surroundings, defeating the purpose of the interface. Additional piloting would be required to engineer a solution of this kind better.

Turn-by-turn builds on LF by also showing landmarks, but it aims to mimic the functionality of most commercial assisted navigation systems. This operation involved explicitly showing the user the direction to follow via a blue line, aligned with Google and Apple Maps colours, and presenting textual navigation cues to further facilitate the process. The goal was to recreate a familiar experience for participants while involving the same underlying cognitive mechanics as in real-world navigation. Creating an effective TBT interface was fundamental to making meaningful comparisons with a different design, such as LF. We now discuss the extent to which we achieved this result.

In-depth research on *human-computer interaction (HCI)* resources, given enough time, would have helped us accurately choose which elements of *user interface (UI)* and *user experience (UX)* to include in our design. The coloured path line in TBT does not fade gradually as the user travels, unlike in other applications. We wonder whether this detail makes the interface look more like a combination of a digital map with a relative location update feature, rather than a turn-by-turn assisted navigation experience. The minimap is not fixed; it rotates with the player's movements, distinguishing our interface from a fixed digital map, since the player does not need to mentally rotate it. The player arrow, being fixed in the centre of the map and pointing north, is also different from the Relative Location Update designed in other studies (Dey et al., 2018; 2019).

One of the most frequently reported opinions among participants concerns directional cues, specifically how unhelpful they were during the tasks. These alerts were designed to be a useful addition to the TBT interface, but in practice, many users completely ignored them during navigation; one participant argued they were even somewhat distracting. The other main feature of TBT could explain this: the guiding path line is arguably already enough help. Its advantage is that it directly associates the minimap perspective with the surroundings, thanks to its level of detail. Furthermore, it already gives a clear direction to follow.

Landmarks' colours were arbitrarily chosen, generally following Google Maps category associations, but without any strategic reasoning. The impact was not further analysed.

5.3.7. Piloting

We acknowledge that a more extensive piloting, given more time, would have allowed us to iteratively refine the design across multiple trials. The original design, described in Section 3.3.3, was only tested by one player, and thus was strongly biased towards the individual skills and preferences of this player: although they could not complete the tasks most likely because of the difficulty, we cannot exclude a coincidental lack of skills; testing with more players would provide certainty of the actual reasons.

After the pilot was unsuccessful, we drastically changed the experimental design. We redrew all the tasks, halving the path length and the number of landmarks, adding a fourth one, and balancing the difficulty differently. Part of the interface design also changed. Landmark indicators on the minimap, represented initially by emojis, are now shown as text. Directional cues in the TBT interface were positioned on the top right of the screen instead of the centre bottom, to better align with commercial navigational systems.

However, we did not pilot again due to a lack of time, and that is also a reason for the drastic design modification. We wanted to make sure the tasks were playable and could be completed, to avoid having to repeat the piloting. After the change, we assumed the experiment would be fine and went on with data collection, being vigilant with our first participant; we were, however, ready to take measures and change the design again if issues arose.

5.3.8. Data processing

Concerning dependent variables, they are logged automatically by Minecraft or, in the case of sketch error, saved to a file. Their parsing, however, was entirely handled manually by us. We did not develop any internal-use script to perform this deterministically and automatically, we relied on hand-transcribed values and measurements. We would start by removing noise lines from the Minecraft log file, eg. system messages that appear during loading; an example can be found in Listing 1. We would then manually go through messages, find those indicating *start of task* and *end of task*, and calculate travel and pointing time subtracting timestamps; we would also note the pointing angle as directly found in the logs. As mentioned in Section 3.5, landmark placement were also manually measured on Photopea, overlapping the correct map sketch with the one completed by the user, and measuring the distance between landmarks with the ruler tool. This process proved not only time-consuming but also more error-prone, as angle values risk being transcribed incorrectly or travel times inaccurately calculated; this is particularly problematic when calculating sketch error, as discussed in Section 5.3.10. To mitigate risks, all values were double-checked once the data collection process was terminated.

5.3.9. Outliers

Our approach with outliers was to keep them in our analyses. They represented a minority of the trials (only 5 out of 128, 3.9%) and were not caused by factors unrelated to the experiment, such as an equipment malfunction. Let us examine them in detail.

All outliers are found when using the landmark-focused interface, representing 7.8% of LF trials (5 out of 64), and possibly indicating a higher susceptibility of this interface to individual differences and thus greater variability. In all cases, the outlying tasks were not the first ones participants played, so that we could exclude a novelty factor; in a few cases, it was the first time they used LF, but it should not matter, as they all went through a testing task at the beginning.

One participant in task 2 picked an angle error out of scale, possibly because it was the most challenging task. Another case presents an abnormal angle time on task 4, phase 1, the last task played by that participant, which might hint at tiredness or frustration.

In 3% of LF trials, participants got lost and arrived at the destination much later. Participant 5 got lost in task 3, phase 2; this is quite unusual, as task 3 is the simplest one. Participant 2 had difficulty finding their way in task 1, phase 1, and task 4, phase 2; this participant has no prior experience with video games, which could partially explain the navigation difficulties. Overall, the numbers are too small to justify a generalised design issue; if the LF interface were poorly designed, we would expect consistent out-of-scale performance when compared to TBT, which is not the case.

Given that these cases are explainable, taking into account individual differences and task design, removing them from our analyses would introduce bias, yielding distributions more similar to our expected results; thus, we decided to keep them.

5.3.10. Variables - landmark placement error

Among the dependent variables measured, sketch accuracy is probably the most problematic methodology-wise.

As described in Section 3.5, the procedure consists of using Photopea to align a participant's filled-in map with the correct landmark locations, then using the ruler tool to measure the Euclidean distance in pixels between the user's attempt and the correct landmark. This method is pragmatic but imprecise for several reasons, each introducing measurement error that accumulates. We manually placed green dots in Photopea to represent the correct landmark positions on the map, based on our perception of their proper positions. When measuring distance with the ruler tool, for each landmark we manually select the centre of the green dot and the centroid of the user-placed labelled rectangle, resulting in approximations in both cases. Furthermore, each distance was calculated only once, rather than repeating sampling and averaging multiple measurements for greater accuracy. Participants were asked to centre the landmarks' labels when placing them on the map to avoid having someone place them next to the correct position, as this would impair our measurement. A more robust pipeline would identify the correct locations as exact in-game coordinates and replicate them on the blank map. It would automate distance calculation, ensuring a consistent measurement method across trials. Nevertheless, this method still captures the desired measure to a reasonable extent. Moderately wrong landmarks are placed about +150px away, and completely wrong ones are placed about +300px; although not accurate, we still obtain a continuous measure of error.

Whenever a player could not recall the position of a landmark, we marked that missing value with -1; then, as we preprocessed sketch data, we replaced every -1 in the dataframe with the largest value for that landmark, indicating the maximum possible error for that landmark. This choice penalises complete landmark omission by treating it as the maximum possible spatial error.

As no other aid is provided, the process requires that a participant notices where a landmark is located on the minimap, understands its name, and recognises it in the virtual environment. This process, however, might not immediately yield an understanding of the landmark's location. We did not test for pure landmark-name association or ask users to recognise landmarks they passed by, showing them pictures as other studies do (Dey et al., 2018; Kapaj et al., 2024).

We did not control for map orientation and landmark list sorting. The former varies depending on the task, since the blank sketch maps are directly taken from a screenshot of a general Minecraft world map; that is just how the task villages are located in the game. We did not consider that when designing the tasks: tasks 1 and 2 travel the map from south to north, while tasks 3 and 4 travel from east to west. A visual representation is available in Figure 5. The landmarks list on the left, from which participants would drag-and-drop labels onto the blank map, would not be in the same order every time. This issue arises during data collection: we manually reset the blank map's initial state (i.e., delete the drawn route line and move the landmarks back to the left) after each sketching iteration, but without preserving the original order; this happened by accident multiple times, and we only realised later in the process.

Although participants were asked to draw their route between the starting point and the end goal, these were not analysed. Apart from the previously mentioned time limitations, there is no consensus method for analysing sketch maps. Some of the considered methods were quite sophisticated (Schwering et al., 2014) and more tailored to real-world maps, so we decided to leave this for future work.

Different strategies and factors can be considered when positioning landmarks. The interface showed the environment from a top-view, rendering colours and details, similar to a satellite view in commercial navigation apps. The blank sketch reported the same map as seen there, just zoomed out to fit entirely. As landmark names were given, one could try to recognise their appearance from the map without actually noticing them during navigation. The best example is the Park in task 4, distinguishable from a 2D top-view map, which might be why it is the landmark with the third-best overall accuracy. This strategy falls short for landmarks with lower top-view visibility (e.g., BnBs). The landmark with the lowest overall accuracy is the Church in task 2. Interestingly, it is the only location not directly on the path; statuary crosses are placed on its rooftop to keep it visible, but players would not necessarily stumble across it during navigation.

5.3.11. Variables - Minecraft and Videogames experience

The scale used to assess the Minecraft and video game experience was chosen for its simplicity. It allows shorter experiment times with fewer questionnaires, and still provides insights about self-reported individual differences. It is, however, a discrete scale ranging from 0 to 2 and captures less complexity than other scales (more on (Norman, 2013)).

6. Conclusions

This study investigated the impact of two different navigation interfaces on spatial knowledge acquisition and spatial navigation performance in virtual environments, comparing a landmark-focused (LF) interface with a turn-by-turn (TBT) interface in Minecraft. Considering methodological limitations, including a small sample size (N=16) and time constraints, the research reported significant exploratory results.

Interface effects on navigation performance The TBT interface effectively reduced learning phase completion times compared to LF, particularly in tasks 1 and 4, confirming that the path line feature improves navigation efficiency during initial route learning. However, the recall phase showed contrasting results: LF users were faster to return in tasks 1 and 2, but this advantage diminished, reaching an equal level in task 3, and then reversed in task 4. This pattern suggests that landmark-focused navigation may support route knowledge acquisition in different ways depending on task characteristics. However, the specific features driving this effect are yet to be investigated. Interface had no significant impact on landmark placement and pointing angle error, suggesting that survey knowledge acquisition is acquired later than route knowledge, as shown in the literature (Siegel & White, 1975).

Task difficulty as the dominant factor Across all dependent variables, except for pointing time, task difficulty emerged as a stronger predictor of performance than interface type. Post-hoc validation confirmed our original difficulty rankings, finding task 3 to be the easiest, task 2 to be the hardest, and tasks 1 and 4 to be at similar levels of challenge. Task 2 consistently showed elevated errors in landmark placement and pointing measures, while task 3 had the lowest learning time and sketch error. These results frame our task design process as correct and motivated.

Individual differences and spatial abilities Correlation analyses revealed that self-reported spatial abilities, specifically mental rotation skills (PTSOT) and experience with video games and Minecraft, are moderate-to-strong predictors of performance across multiple measures. Video game and Minecraft experience showed strong negative correlations with completion times and pointing error, while PTSOT showed mildly strong positive correlations with angle error and landmark placement error. Notably, the landmark-focused interface was the only one with outlying values, and, in general, models improved when random slopes for participants were added, suggesting greater performance variability across subjects. These results hint that individual differences moderate interface effectiveness, in line with their importance in the literature.

Gaze Although we did not go into detail with eye tracking measures, a first exploration revealed a larger amount of fixations, especially on the interface, on landmark-focused trials. This result is in contrast with our initial Hypothesis H2, where we expected users to pay less attention to the environment when using the turn-by-turn interface. This outcome is likely related to the higher engagement required to navigate with only landmarks, and it suggests that more investigation is required in this direction.

6.1. Future Perspectives

This thesis presents exploratory, empirical results on how a landmark-focused interface can be implemented in a Minecraft virtual environment and how it impacts navigation compared to a turn-by-turn system. However, the extent largely depends on the task's and environment's intrinsic characteristics. This finding suggests that future research should investigate design features such as visual distinctiveness, route complexity, and landmark salience to understand which contexts benefit most from a landmark-focused design.

As navigational cues were mostly ignored in turn-by-turn navigation, we ask if there is any potential value to the concept or if virtual environment navigation does not benefit from them as

much as car driving. Future studies could explore different display locations or visualisation styles to further understand our results.
This study's short-term, one-shot experimental design does not address the longer-term cognitive costs of habitual GPS use seen in the literature. A longitudinal study would be more appropriate to track repeated exposure to different interfaces and better understand how our landmark-focused design may affect spatial learning in the long term.

Considering the results we obtained and assuming greater time and monetary compensation for participants, our experimental design would follow slightly different strategies. We believe that further research and experimentation would be crucial to more accurately inform our design decisions.
First, we would aim to recruit a larger number of individuals across different age groups and introduce a more detailed video game experience questionnaire, as we observed a strong correlation. The task design process would be more research-based, and we would redesign the tasks, being more conscious about the features to include in our difficulty scale calculation. Changing the interfaces would be trickier, as it would require a deeper understanding of Minecraft APIs and Java, and how Xaero's ecosystem works; nevertheless, it would be interesting to try different variations in minimap size, different locations of the screen, or having the user manually toggle the interface by pressing a key to study their reliance on it. The entire process would be aided by additional piloting sessions to iteratively assess our design's strengths and weaknesses. A better assignment strategy would be used to control for bias, and, incidentally, a higher number of participants would allow us to make more advanced use of statistics. Finally, more focus would be placed on analysing eye-tracking data, as we recognise a valuable opportunity that has not been fully explored.

6.2. Concluding Thoughts

While this study cannot definitively confirm whether landmark-focused interfaces promote enhanced spatial knowledge acquisition and navigation performance across different contexts in virtual environments, it provides preliminary hints that interface design matters and that task characteristics and individual differences substantially influence its effects. The dominance of task difficulty effects suggests that environmental design may be as important as interface design for supporting spatial learning in virtual and real-world navigation contexts.

Despite limitations, this study establishes a foundation for understanding how minimalist, landmark-focused interfaces function in virtual environments. It opens the way for bridging research into real-world applications, to developing navigation aids that balance efficiency with spatial knowledge acquisition, a growingly important consideration as GPS navigation systems become increasingly ubiquitous in daily life.

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A Appendix

A.1 Additional dataframe visualisations

Here is a reference image for the structure of each dataframe used in the analysis, discussed in Section 3.5.

			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
task	landmark	on_turn																
1	bnb	True	58	167	33	176	915	78	267	353	173	13	58	34	61	242	198	331
	pub	False	20	177	51	115	87	77	92	182	226	8	200	8	244	81	244	163
	fish shop	True	75	41	105	101	456	57	214	459	576	41	135	78	131	892	892	459

Figure 24: A snippet from `sketch` DataFrame. Each row represents a landmark on a given task, reports whether it is on an intersection, and shows the placement error for each participant, numbered from 1 to 16.

	task1	iface1	task2	iface2	task3	iface3	task4	iface4	sbsod	ptsot	minecraft	videogames
name												
1	1	tbt	2	lf	3	tbt	4	lf	3.74	7.20	0	1
2	4	tbt	1	lf	2	lf	3	tbt	3.74	18.00	2	[illegible]
3	3	tbt	4	lf	1	lf	2	tbt	5.73	26.27	0	0

Figure 25: A snippet from `general` DataFrame. Each row contains some general data for participants, such as their SBSOD score, and the order in which they played the tasks.

				tot_dur_fixations	avg_dur_fixations	n_fixations	perc_dur_fixations	perc_n_fixations
name	task	aoi	iface					
1	1	direction	tbt	0	0.0	0	0.000	0.000
		env	tbt	42487	256.0	166	68.167	72.807
		minimap	tbt	19841	320.0	62	31.833	27.193

Figure 26: A snippet from `fixations` DataFrame. Each row contains fixation information for a given participant, task, phase, and Area of Interest.

A.2 Aggregated analysis

As mentioned in Section 3.5, here are the results from the aggregated analysis. This approach allowed a straightforward comparison between the two interfaces, treating them as independent conditions on a single task. Given the non-normality of the distributions (tested with the Shapiro-Wilk test) and the small sample size, we use the Mann-Whitney U test for significance. The process of squashing the tasks together was done in two different ways. A more naïve one that groups the data by interface, and a more accurate one that groups by interface, while also weighting by task difficulty.

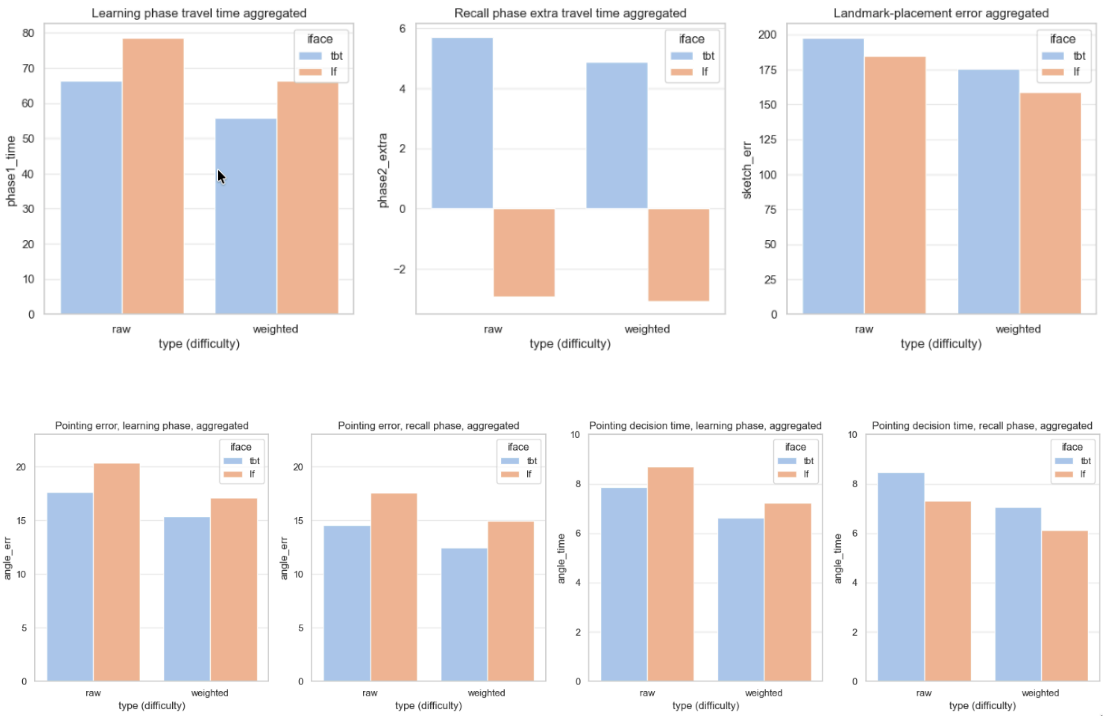

Figure 27: Aggregated exploratory analysis.

A.3 Diagnostic plots

Here we report the entire set of diagnostic plots for LMMs, introduced in Section 4.2 with **learning phase travel time**; we include the latter one for consistency. We also report the full model outputs.

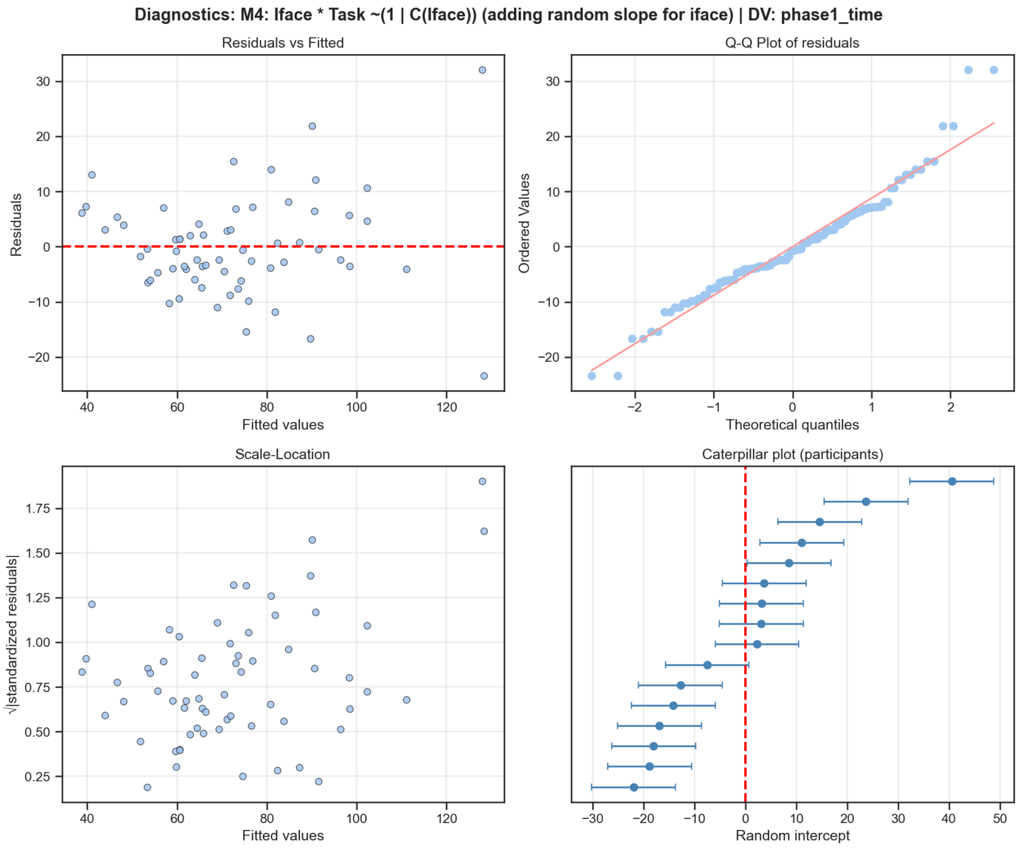

Figure 30: Diagnostic charts regarding the fitting process, **learning phase travel time** variable.

```
FINAL MODEL SUMMARY (REML)
======================================================================
                   Mixed Linear Model Regression Results
===========================================================================
Model:                  MixedLM       Dependent Variable:        phase1_time
No. Observations:       128           Method:                    REML
No. Groups:             16            Scale:                     96.5052
Min. group size:        8             Log-Likelihood:            -478.1806
Max. group size:        8             Converged:                 Yes
Mean group size:        8.0
---------------------------------------------------------------------------
                              Coef.   Std.Err.   z    P>|z|  [0.025  0.975]
---------------------------------------------------------------------------
Intercept                      87.422    5.263 16.611 0.000  77.106  97.737
C(iface)[T.tbt]               -16.628    4.402 -3.778 0.000 -25.255  -8.001
C(task)[T.2]                   -8.758    4.018 -2.180 0.029 -16.634  -0.882
C(task)[T.3]                  -26.566    4.013 -6.620 0.000 -34.430 -18.701
C(task)[T.4]                    0.387    4.130  0.094 0.925  -7.707   8.482
C(iface)[T.tbt]:C(task)[T.2]   11.575    5.842  1.981 0.048   0.125  23.026
C(iface)[T.tbt]:C(task)[T.3]    6.941    5.861  1.184 0.236  -4.546  18.428
C(iface)[T.tbt]:C(task)[T.4]   -1.503    6.074 -0.247 0.805 -13.407  10.402
Group Var                     320.730   13.893
Group x C(iface)[T.tbt] Cov  -122.303    6.659
```

```
C(iface)[T.tbt] Var          51.551    4.059
==========================================================================
```

```
======================================================================
WALD JOINT TESTS (OMNIBUS)
======================================================================
- Interface: <Wald test (chi2): statistic=[[18.76477868]], p-
value=0.0003057865451509805, df_denom=3>
- Task: <Wald test (chi2): statistic=[[60.40302156]], p-
value=4.821023725876845e-13, df_denom=3>
- Interaction Interface x Task: <Wald test (chi2): statistic=[[6.9965585]], p-
value=0.07200754558066728, df_denom=3>
```

```
======================================================================
SIMPLE EFFECTS
======================================================================
         Task  Coefficient        SE         z   p-value  sig
0  Task 1 (ref)   -16.628287  4.401639 -3.777748  0.000158  ***
1        Task 2    -5.053069  4.403120 -1.147611  0.251129   ns
2        Task 3    -9.687618  4.422429 -2.190565  0.028483    *
3        Task 4   -18.131026  4.411075 -4.110342  0.000040  ***
```

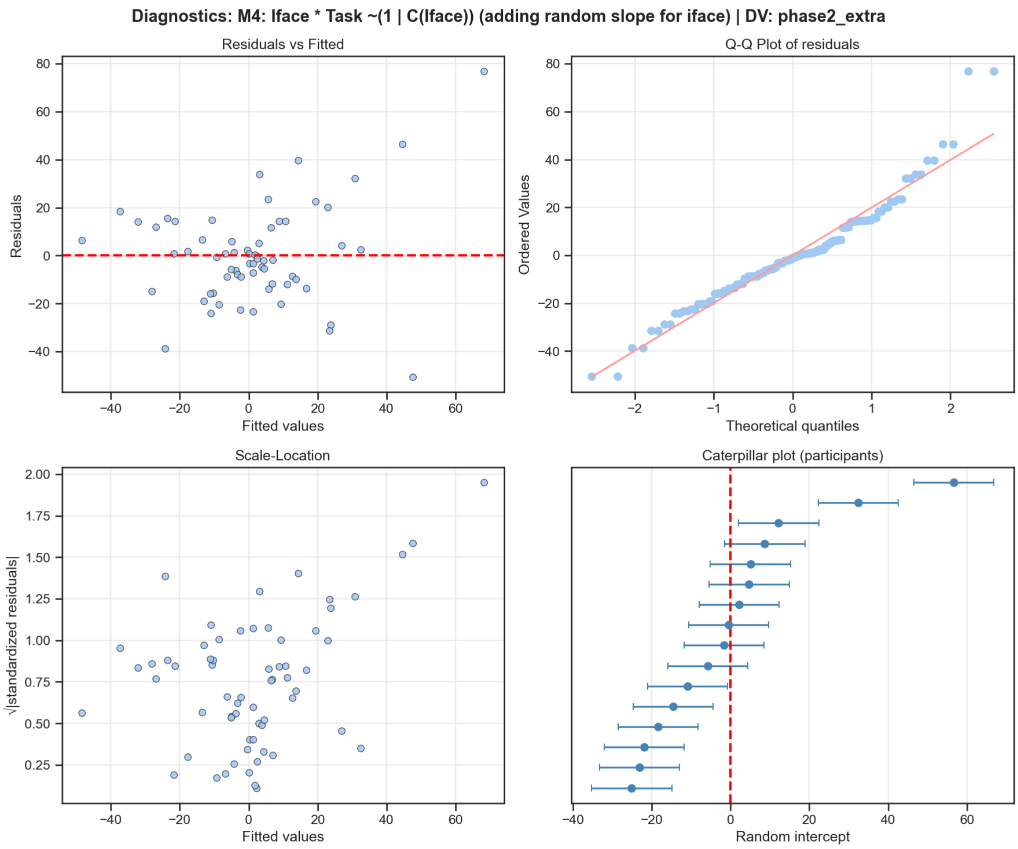

Figure 31: Diagnostic charts regarding the fitting process, **recall phase travel time** variable.

```
======================================================================
                 Mixed Linear Model Regression Results
==========================================================================
Model:                   MixedLM      Dependent Variable:      phase2_extra
No. Observations:        128          Method:                  REML
No. Groups:              16           Scale:                   524.9195
```

```
Min. group size:         8            Log-Likelihood:          -576.0906
Max. group size:         8            Converged:               Yes
Mean group size:         8.0
---------------------------------------------------------------------------
                              Coef.   Std.Err.   z    P>|z|  [0.025  0.975]
---------------------------------------------------------------------------
Intercept                    -26.455    8.831 -2.996 0.003 -43.763  -9.147
C(iface)[T.tbt]               35.081   11.297  3.105 0.002  12.939  57.223
C(task)[T.2]                  17.416    9.507  1.832 0.067  -1.217  36.050
C(task)[T.3]                  38.048    9.414  4.042 0.000  19.598  56.498
C(task)[T.4]                  38.605    9.751  3.959 0.000  19.494  57.717
C(iface)[T.tbt]:C(task)[T.2]  -21.368   13.166 -1.623 0.105 -47.173   4.437
C(iface)[T.tbt]:C(task)[T.3]  -40.047   13.112 -3.054 0.002 -65.746 -14.348
C(iface)[T.tbt]:C(task)[T.4]  -44.285   13.505 -3.279 0.001 -70.755 -17.815
Group Var                    573.700   12.429
Group x C(iface)[T.tbt] Cov  -548.941   13.320
C(iface)[T.tbt] Var          734.147   17.701
===========================================================================

======================================================================
WALD JOINT TESTS (OMNIBUS)
======================================================================
- Interface: <Wald test (chi2): statistic=[[13.11050694]], p-
value=0.004403621239759918, df_denom=3>
- Task: <Wald test (chi2): statistic=[[21.86263442]], p-
value=6.966795136068958e-05, df_denom=3>
- Interaction Interface x Task: <Wald test (chi2): statistic=[[13.62325961]], p-
value=0.0034655074494857317, df_denom=3>

======================================================================
SIMPLE EFFECTS
======================================================================
           Task  Coefficient         SE         z   p-value sig
0  Task 1 (ref)    35.081095  11.297233  3.105282  0.001901  **
1        Task 2    13.713448  11.385427  1.204474  0.228407  ns
2        Task 3    -4.965979  11.342800 -0.437809  0.661525  ns
3        Task 4    -9.203564  11.381571 -0.808637  0.418724  ns
```

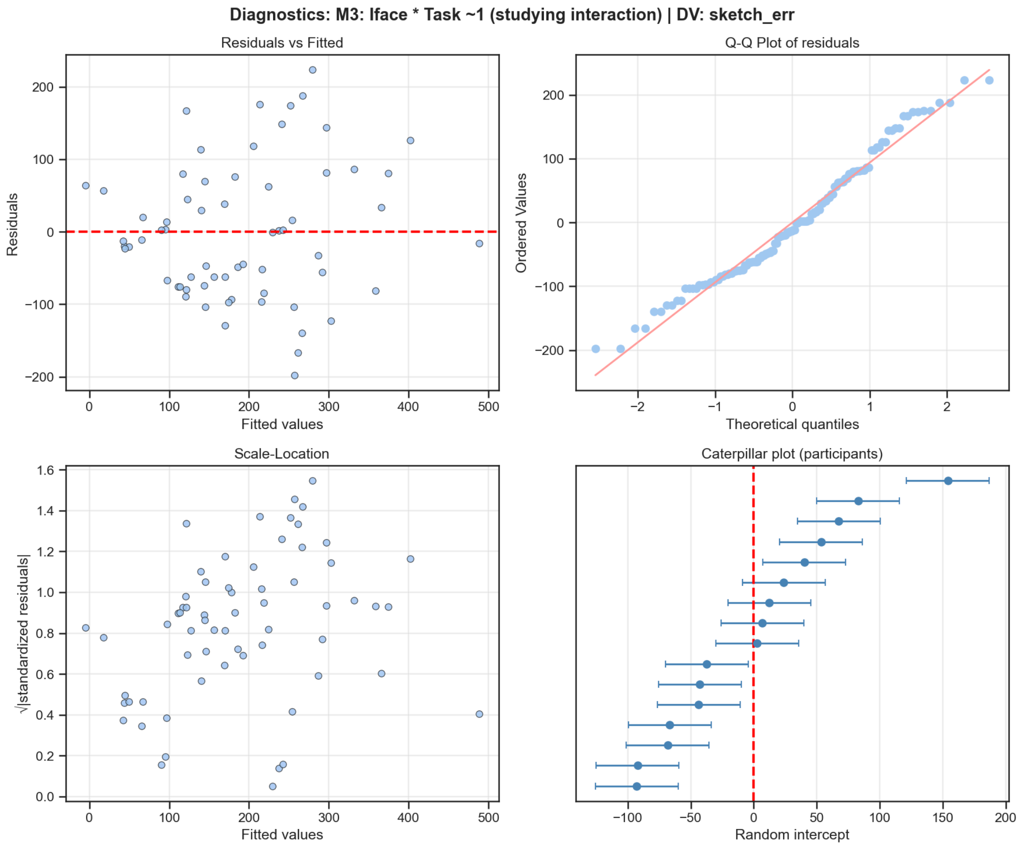

Figure 32: Diagnostic charts regarding the fitting process, **landmark placement error** variable.

```
======================================================================
                 Mixed Linear Model Regression Results
=============================================================================
Model:                    MixedLM      Dependent Variable:       sketch_err
No. Observations:         128          Method:                   REML
No. Groups:               16           Scale:                    10345.7558
Min. group size:          8            Log-Likelihood:           -748.4383
Max. group size:          8            Converged:                Yes
Mean group size:          8.0
-----------------------------------------------------------------------------
                              Coef.   Std.Err.   z     P>|z|  [0.025   0.975]
-----------------------------------------------------------------------------
Intercept                     189.535   33.400  5.675 0.000  124.071 254.999
C(iface)[T.tbt]               -56.966   41.078 -1.387 0.166 -137.476  23.545
C(task)[T.2]                   59.737   39.025  1.531 0.126  -16.750 136.224
C(task)[T.3]                 -102.441   39.326 -2.605 0.009 -179.519 -25.362
C(task)[T.4]                   22.543   39.868  0.565 0.572  -55.597 100.683
C(iface)[T.tbt]:C(task)[T.2]  142.296   59.205  2.403 0.016   26.256 258.337
C(iface)[T.tbt]:C(task)[T.3]   80.371   59.999  1.340 0.180  -37.225 197.966
C(iface)[T.tbt]:C(task)[T.4]   58.727   61.411  0.956 0.339  -61.637 179.091
Group Var                    5927.072   28.096
=============================================================================
```

```
======================================================================
WALD JOINT TESTS (OMNIBUS)
======================================================================
- Interface: <Wald test (chi2): statistic=[[1.92315357]], p-
value=0.16550942925904874, df_denom=1>
```

```
- Task: <Wald test (chi2): statistic=[[18.25640022]], p-
value=0.0003894089850035035, df_denom=3>
- Interaction Interface x Task: <Wald test (chi2): statistic=[[6.05286595]], p-
value=0.10906631883870187, df_denom=3>

======================================================================
SIMPLE EFFECTS
======================================================================
          Task  Coefficient         SE         z   p-value sig
0  Task 1 (ref)   -56.965596  41.077657 -1.386778  0.165509  ns
1        Task 2    85.330683  40.830112  2.089896  0.036627   *
2        Task 3    23.405135  41.152432  0.568742  0.569531  ns
3        Task 4     1.761027  40.809772  0.043152  0.965580  ns
```

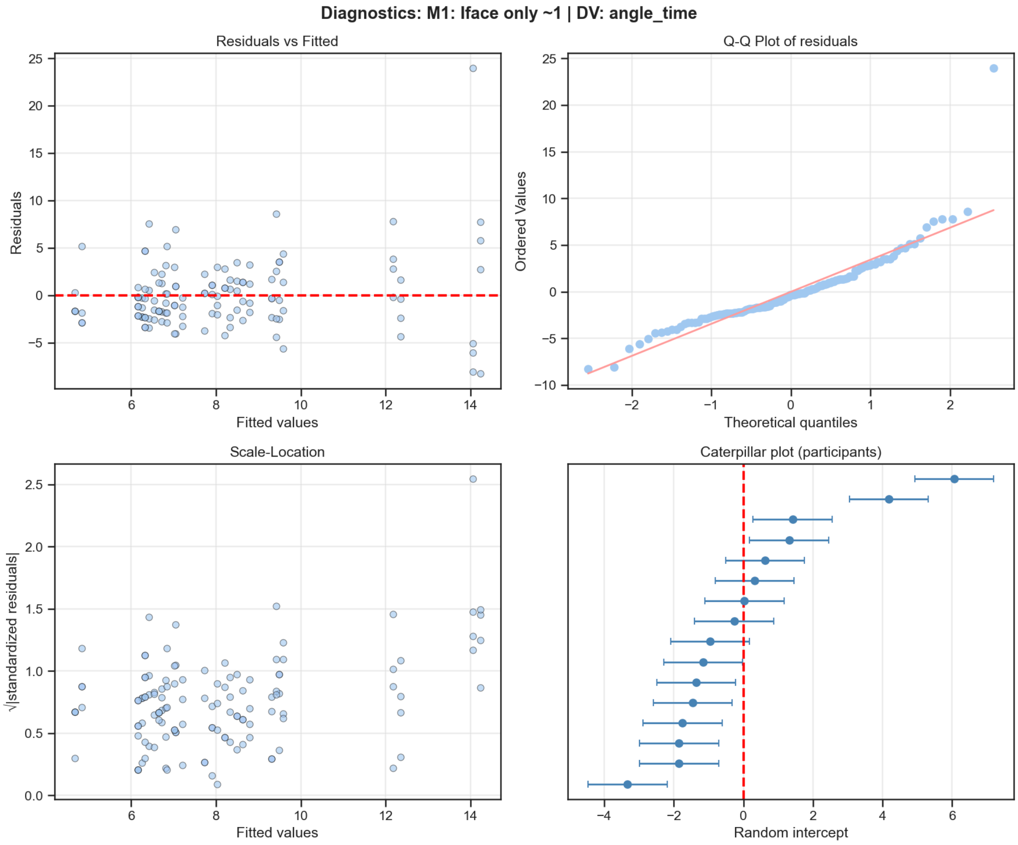

Figure 33: Diagnostic charts regarding the fitting process, **pointing decision time** variable.

```
======================================================================
         Mixed Linear Model Regression Results
========================================================
Model:             MixedLM Dependent Variable: angle_time
No. Observations: 128     Method:             REML
No. Groups:        16      Scale:              15.3056
Min. group size:   8       Log-Likelihood:     -366.5481
Max. group size:   8       Converged:          Yes
Mean group size:   8.0
--------------------------------------------------------
                Coef. Std.Err.   z   P>|z| [0.025 0.975]
--------------------------------------------------------
Intercept       8.000    0.831 9.627 0.000  6.371  9.629
C(iface)[T.tbt] 0.172    0.692 0.249 0.804 -1.184  1.527
Group Var       7.222    0.908
```

```
========================================================
```

```
======================================================================
WALD JOINT TESTS (OMNIBUS)
======================================================================
- Interface: <Wald test (chi2): statistic=[[0.06176233]], p-
value=0.8037318764307265, df_denom=1>
```

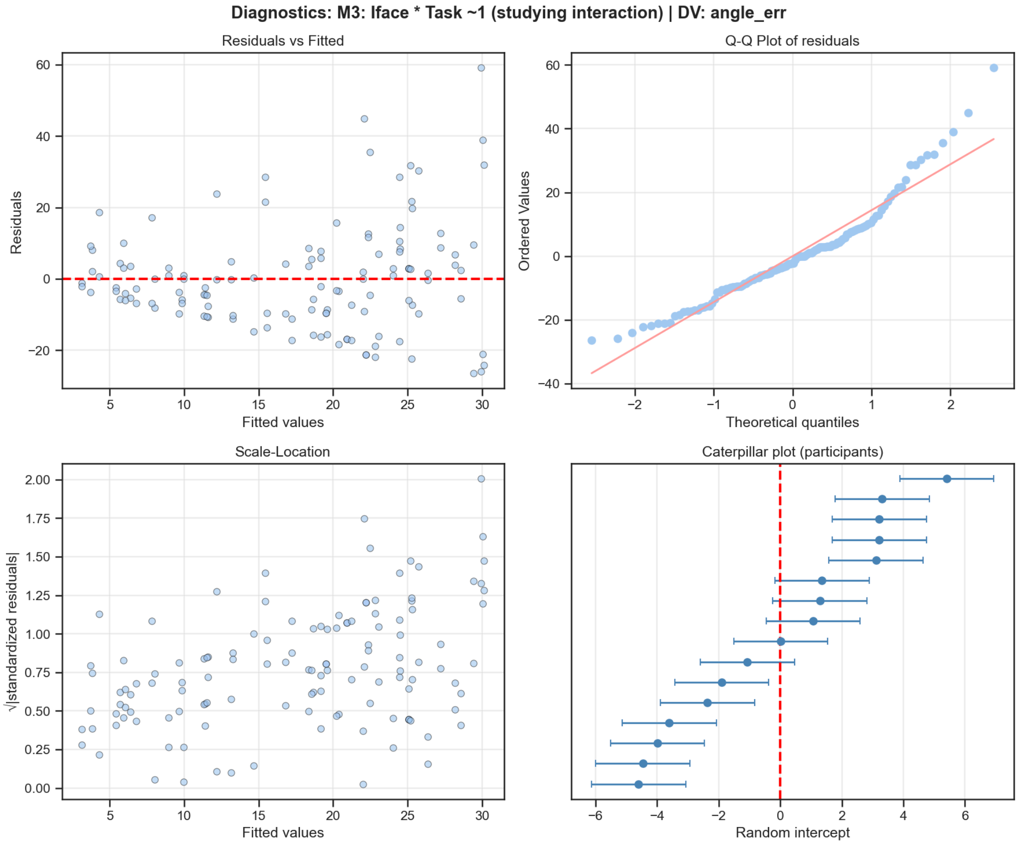

Figure 34: Diagnostic charts regarding the fitting process, **pointing angle error** variable.

```
======================================================================
                  Mixed Linear Model Regression Results
=========================================================================
Model:                  MixedLM        Dependent Variable:        angle_err
No. Observations:       128            Method:                    REML
No. Groups:             16             Scale:                     242.5542
Min. group size:        8              Log-Likelihood:            -515.0174
Max. group size:        8              Converged:                 Yes
Mean group size:        8.0
-------------------------------------------------------------------------
                              Coef.  Std.Err.   z    P>|z|  [0.025 0.975]
-------------------------------------------------------------------------
Intercept                      6.758    4.263  1.585 0.113  -1.597 15.113
C(iface)[T.tbt]                1.546    6.023  0.257 0.797 -10.258 13.350
C(task)[T.2]                  20.057    5.777  3.472 0.001   8.734 31.379
C(task)[T.3]                  12.396    5.821  2.130 0.033   0.987 23.804
C(task)[T.4]                  16.390    5.861  2.797 0.005   4.904 27.877
C(iface)[T.tbt]:C(task)[T.2]  -4.364    8.535 -0.511 0.609 -21.093 12.365
C(iface)[T.tbt]:C(task)[T.3] -10.666    8.654 -1.233 0.218 -27.627  6.294
C(iface)[T.tbt]:C(task)[T.4]  -2.718    8.760 -0.310 0.756 -19.888 14.452
Group Var                     24.360    1.422
```

```
=========================================================================

======================================================================
WALD JOINT TESTS (OMNIBUS)
======================================================================
- Interface: <Wald test (chi2): statistic=[[0.06592429]], p-
value=0.7973661723955453, df_denom=1>
- Task: <Wald test (chi2): statistic=[[13.43912912]], p-
value=0.0037770893288520176, df_denom=3>
- Interaction Interface x Task: <Wald test (chi2): statistic=[[1.69028771]], p-
value=0.6390952134886434, df_denom=3>

======================================================================
SIMPLE EFFECTS
======================================================================
           Task  Coefficient        SE         z   p-value sig
0  Task 1 (ref)     1.546352  6.022623  0.256757  0.797366  ns
1        Task 2    -2.817399  5.858847 -0.480879  0.630602  ns
2        Task 3    -9.120061  5.897810 -1.546347  0.122021  ns
3        Task 4    -1.171392  5.886616 -0.198992  0.842269  ns
```

A.4 Unmasked correlations

Here we present the unmasked heatmap discussed in Figure 18 regarding correlations between individual skills.

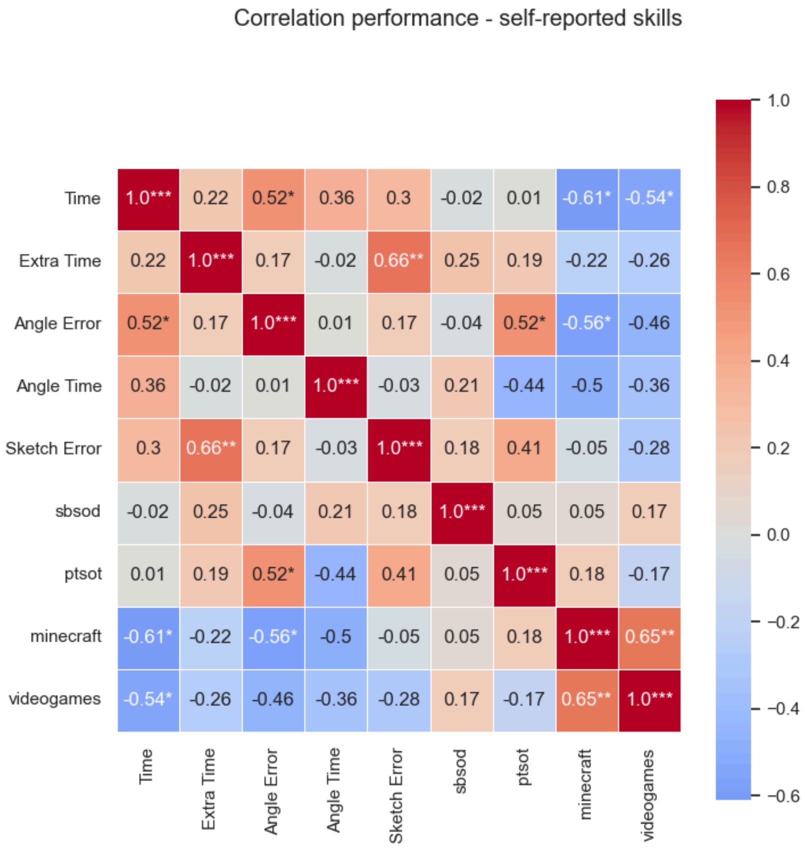

Figure 35: Correlations in-between dependent variables, including individual skills and tracked performance.

	Time	Extra Time	Angle Error	Angle Time	Sketch Error	sbsod	ptsot	minec.	v.g.
Time	0.000	0.408	0.040	0.168	0.254	0.952	0.978	0.012	0.030
Extra Time	0.408	0.000	0.520	0.935	0.005	0.355	0.476	0.403	0.335
Angle Error	0.040	0.520	0.000	0.970	0.528	0.888	0.039	0.023	0.072
Angle Time	0.168	0.935	0.970	0.000	0.901	0.426	0.089	0.050	0.177
Sketch Error	0.254	0.005	0.528	0.901	0.000	0.498	0.111	0.846	0.300
sbsod	0.952	0.355	0.888	0.426	0.498	0.000	0.841	0.860	0.526
ptsot	0.978	0.476	0.039	0.089	0.111	0.841	0.000	0.512	0.540
minec.	0.012	0.403	0.023	0.050	0.846	0.860	0.512	0.000	0.007
v.g.	0.030	0.335	0.072	0.177	0.300	0.526	0.540	0.007	0.000

Table 15: p-values for the correlation matrix between variables.

A.5 Gaze

```
Models:
m0: n_fixations ~ 1 + (1 | name)
m1: n_fixations ~ iface + task + aoi + (1 | name)
m2: n_fixations ~ (iface + task + aoi)^2 + (1 | name)
m3: n_fixations ~ iface * task * aoi + (1 | name)
   npar    AIC    BIC   logLik -2*log(L)    Chisq Df Pr(>Chisq)
m0    2 2731.1 2736.8 -1363.55    2727.1
m1    7 2025.3 2045.2 -1005.63    2011.3 715.8343  5  < 2.2e-16 ***
m2   14 1958.6 1998.5  -965.31    1930.6  80.6511  7  1.015e-14 ***
m3   17 1957.3 2005.8  -961.67    1923.3   7.2759  3     0.0636 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Listing 2: Likelihood Ratio Test output from comparing the different models for fixation number. M2 is the chosen model.

```
Warning message:
In checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv,  :
  Model failed to converge with max|grad| = 0.00785928 (tol = 0.002, component 1)

Warning message:
In checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv,  :
  Model failed to converge with max|grad| = 0.0540492 (tol = 0.002, component 1)

  Models:
  mdur0: avg_dur_fixations ~ 1 + (1 | name)
  mdur1: avg_dur_fixations ~ iface + task + aoi + (1 | name)
  mdur2: avg_dur_fixations ~ iface + task + aoi + iface:task + iface:aoi +
task:aoi + (1 | name)
  mdur3: avg_dur_fixations ~ iface * task * aoi + (1 | name)
        npar    AIC    BIC  logLik -2*log(L)   Chisq Df Pr(>Chisq)
  mdur0    3 1285.1 1293.5 -639.54    1279.1
  mdur1    8 1273.3 1295.8 -628.67    1257.3 21.7440  5  0.0005856 ***
  mdur2   15 1285.8 1327.8 -627.89    1255.8  1.5681  7  0.9798566
  mdur3   18 1291.0 1341.5 -627.49    1255.0  0.7947  3  0.8507333
  ---
  Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Listing 3: Likelihood Ratio Test output from comparing the different models for fixation duration. M1 is the chosen model.

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: n_fixations
               Chisq Df Pr(>Chisq)
(Intercept) 423.4040  1  < 2.2e-16 ***
iface         1.9034  1    0.16770
task         44.1449  3  1.406e-09 ***
aoi         164.9804  1  < 2.2e-16 ***
iface:task    9.5534  3    0.02277 *
iface:aoi    16.9147  1  3.910e-05 ***
task:aoi     50.7125  3  5.633e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Listing 4: Wald’s test on M2 testing for interface, task, and AOI effects, and their interactions.

```
emmeans(m2, ~ iface | aoi, type="response")

== Summary ==
aoi = env:
 iface rate    SE  df asymp.LCL asymp.UCL
 lf    84.8 18.55 Inf      55.2     130.2
 tbt   74.0 16.21 Inf      48.2     113.7

aoi = minimap:
 iface rate    SE  df asymp.LCL asymp.UCL
 lf    59.9 13.12 Inf      39.0      92.0
 tbt   44.3  9.73 Inf      28.8      68.1

Results are averaged over the levels of: task
Confidence level used: 0.95
Intervals are back-transformed from the log scale

== Pairs ==
aoi = env:
 contrast ratio     SE  df z.ratio p.value
 lf / tbt  1.15 0.0289 Inf 5.365   <.0001

aoi = minimap:
 contrast ratio     SE  df z.ratio p.value
 lf / tbt  1.35 0.0424 Inf 9.581   <.0001

Results are averaged over the levels of: task
Tests are performed on the log scale
```
Listing 5: Effect of interface on AOI, output from R `emmeans`.